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Economy

Will younger-gen spending hit a gas-price speed bump?

24 March 2026

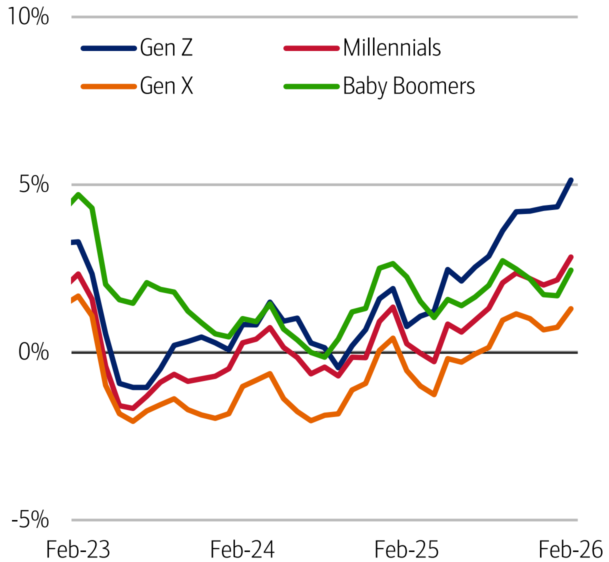
Key takeaways

- Gen Z and Millennials have seen an improvement in their spending growth over the last year: in Bank of America credit and debit card data, younger generations' spending growth was higher than older generations as of February 2026.
- What's driving this? In our view, easing rent pressures are a key factor. Younger consumers are seeing rent payment growth below their wage growth according to our data. As a result their spending on discretionary items such as electronics, clothing and restaurants has improved. Tax refunds are an additional tailwind.
- But the current oil shock poses risks to this picture. Younger generations' gasoline spending is relatively high compared to their discretionary spending, so there is the potential they will need to pullback most aggressively in the face of higher gasoline prices. Further out, while the overall labor market may be "low-hire, low-fire", it poses particular challenges for Gen Z, with potential knock-on headwinds to their spending.

Gen Z and Millennial households' spending have surged in the past year...

Gen Z and Millennial households' spending growth has markedly improved over the past year, especially compared to Gen X and Baby Boomers. In fact, Bank of America aggregated credit and debit card data indicates that Gen Z's year-over-year (YoY) spending growth surpassed that of Baby Boomers in mid-2025, after trailing for nearly two years (Exhibit 1). More recently, Millennials' spending growth surpassed older generations' in December 2025, after lagging for nearly three years.

Exhibit 1: Gen Z and Millennial spending growth has surged...
Total credit and debit card spending per household, according to Bank of America card data, by household generations (3-month moving average, YoY%, non-seasonally adjusted (NSA))



Source: Bank of America internal data

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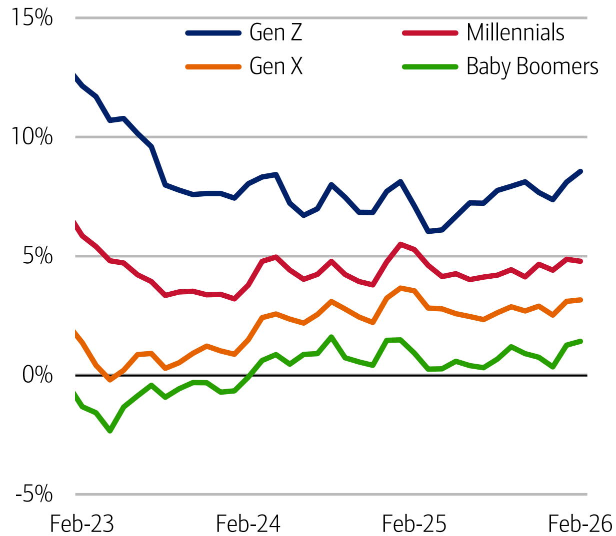
Exhibit 2: ...likely supported by solid wage growth for these generations
After-tax wage and salary growth by household generations, based on Bank of America aggregated consumer deposit account data (3-month moving average, YoY%, seasonally adjusted (SA))



Source: Bank of America internal data

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Part of the reason for this recent uptick is likely the continued strength in younger generations' after-tax wage growth. Bank of America data suggests that even though wage growth has cooled from the peaks seen three years ago, it remains fairly strong, up around 9% and 5% YoY for Gen Z and Millennials, respectively (Exhibit 2). However, Millennials do not appear to have experienced the sort of wage growth acceleration that, on its own, would support the recent surge in spending growth.

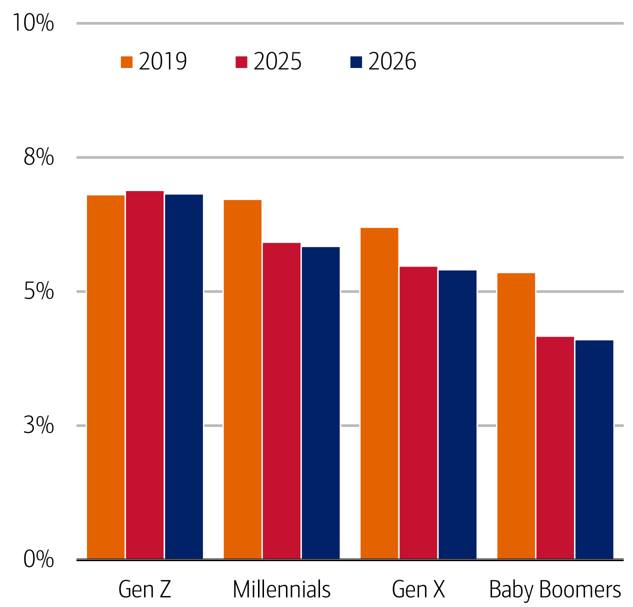
...but will rising oil prices spoil the party?

A hot topic for everyone right now is how rising gas prices, due to the conflict in Iran, might impact both the consumer and the wider economy. The average national gas price is up around 26% YoY, according to daily data from American Automobile Association (AAA) as of March 23rd.

In our view, the discretionary spending of younger generations may be most susceptible to rising gas prices. Before the recent crisis, most generations had seen their gasoline spending, as a share of total card spending, fall compared to pre-pandemic levels. But Gen Z's remained relatively high and steady, while Millennials' share was above that of older generations (Exhibit 3). In our view, Gen Z's elevated share likely reflects them entering the workforce and experiencing the daily (or mostly daily) commute for the first time, while their incomes are still relatively low. Meanwhile, Millennials are, in our view, likely the generation doing the most driving, given their responsibilities with work and families.

Exhibit 3: Gen Z appears most exposed to rising gas prices compared to other generations

Gasoline spending as a share of total card spending per household and by generation (12-month rolling average, %)



Source: Bank of America internal data

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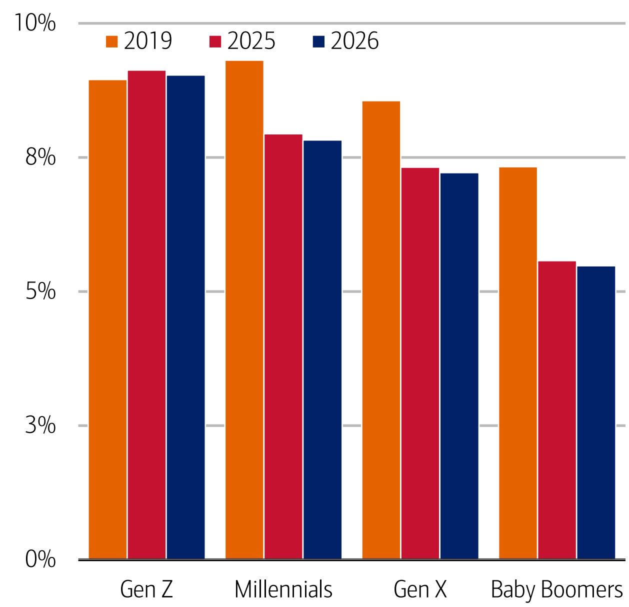
Exhibit 4: Younger generations' discretionary spending is most at risk

Gasoline spending relative to discretionary spending per household and by generation (12-month rolling average, %)



Source: Bank of America internal data.

Note: discretionary spending includes all card spending except gas, groceries and utilities.

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A similar but more pronounced pattern exists across the generations for discretionary spending (i.e., travel, electronics, restaurants) (Exhibit 4). Even though Gen Z households spend less on both gasoline and discretionary items in dollar terms compared to other generations, their outlay on gasoline is highest relative to their discretionary spending. And Millennials spend more on gasoline compared to discretionary spending than Gen X or Baby Boomers. So, both Gen Z and Millennials may be even more prone to cutting back on this "nice-to-have" spending amid higher gasoline prices.

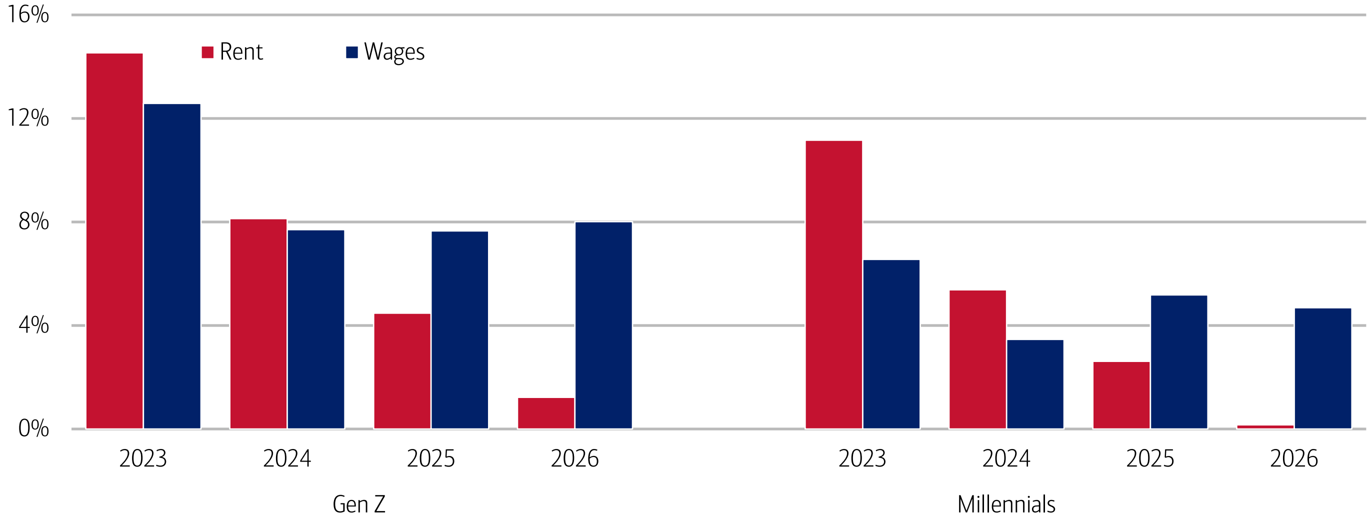
Note that the spending data is focused here on Gen Z *households*, who are less likely to be living at home with their parents/other relatives. Reflecting this, the Gen Z consumers we analyze are likely to be older than the overall cohort, while Gen Zers still living with parents may well experience less pressure on their budgets from gasoline.

Accelerating spending growth has been aided by decelerating rent growth

However, there is some "good news" that may help younger generations ride out the impact of higher gas prices – rent growth is cooling. In Bank of America data, median rent growth for Gen Z and Millennials slowed significantly in the 12 months to February 2026 compared to 2024 and 2023. And, notably, wage growth for these generations is currently outpacing rent payment growth (Exhibit 5).

Exhibit 5: Rental payment growth has slowed relative to wage growth for Millennials and Gen Z

Median rent payments and average after-tax wages, based on Bank of America internal data, for Gen Z and Millennials (yearly 3-month moving average to February, YoY%)



Source: Bank of America internal data

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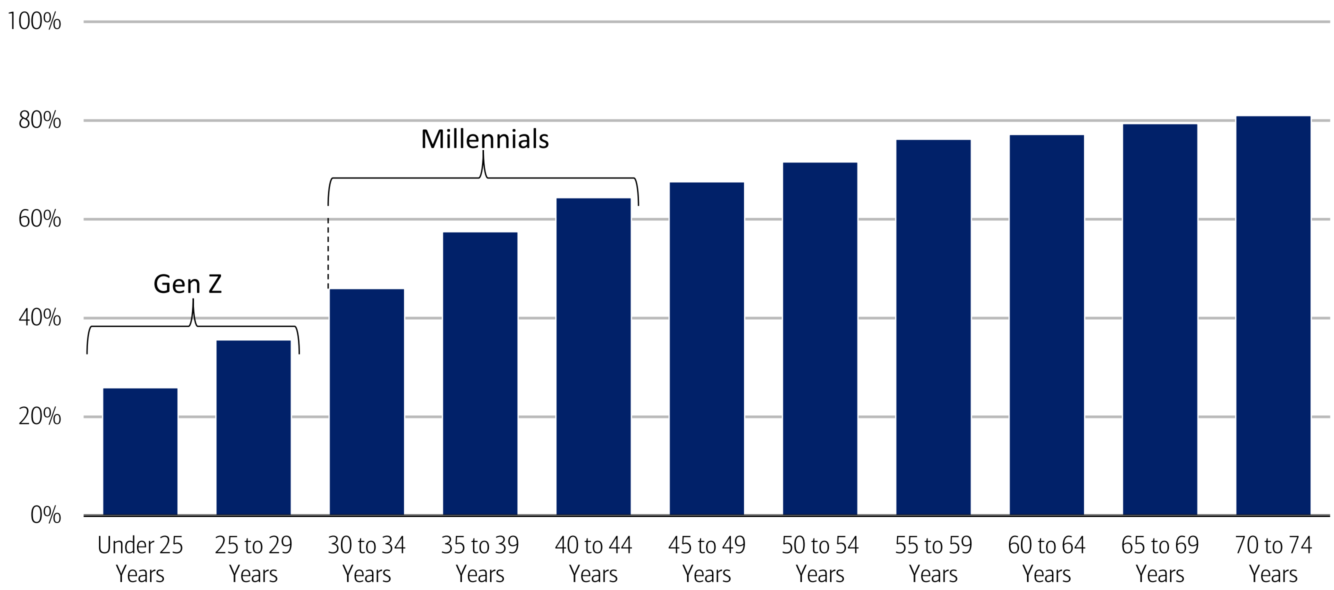
Cooling rent growth disproportionately benefits the young

Given that housing often makes up the largest share of a household's expenses, cooling rent growth could give some younger consumers a cushion to offset part of the increase in gasoline prices. Additionally, slowing increases in rent are especially important for younger generations, considering many are renters, whereas older generations are more likely to be homeowners (Exhibit 6).

Exhibit 6: Younger generations have lower homeownership rates

US homeownership rates by age range (fourth quarter 2025, %)



Source: US Census Bureau

Note: Age generations are approximate and may not perfectly match age ranges.

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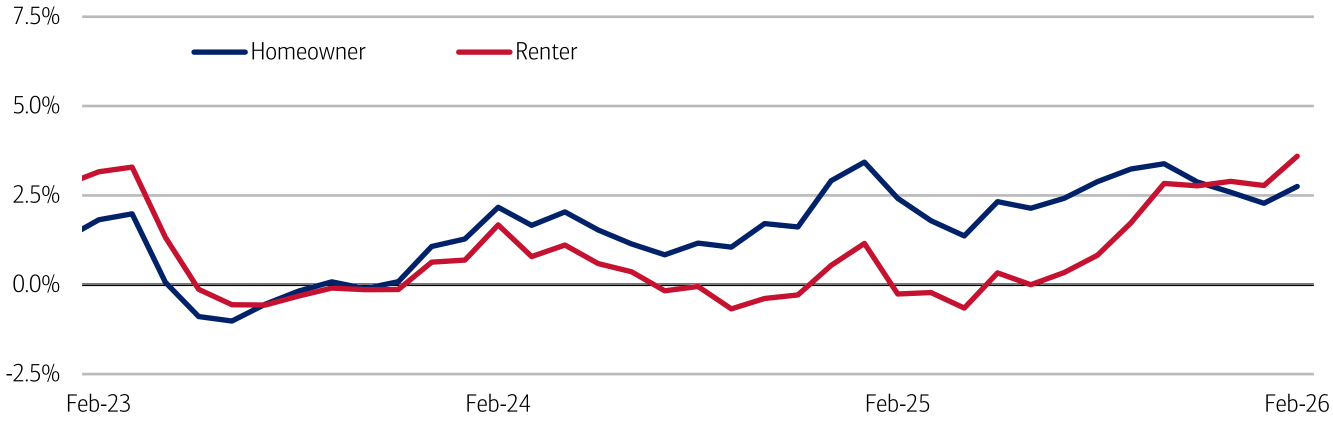
In fact, Bank of America card data shows that while Gen Z and Millennial homeowners' spending growth has been solid for the past two years, it is renters in these generations whose spending growth that has accelerated considerably over the past seven months (Exhibit 7).

Exhibit 7: Renters have accelerated their spending considerably over nearly the past year

Total card spending per household for Gen Z and Millennials, based on Bank of America data, split by homeowners and renters (3-month moving average, YoY%)



Source: Bank of America internal data

Note: "Homeowners" include households who both own and rent concurrently.

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Easing rental pressures gives discretionary spending a boost

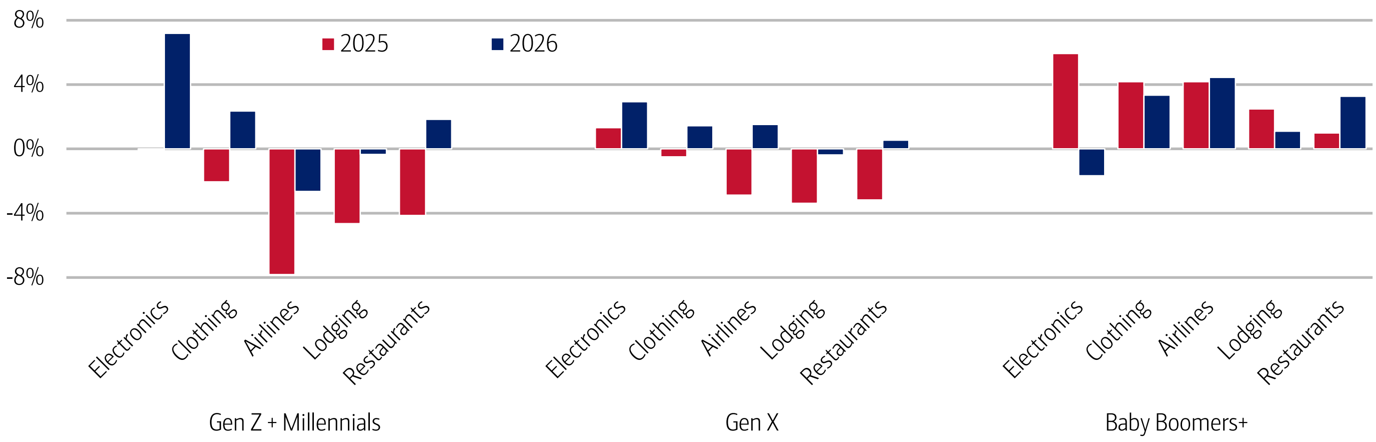
So, for Gen Z and Millennials in particular, easing rental payments and slowing inflation pressures on other necessity spending (including previously in gas prices), has allowed them to ramp up their discretionary spending over the past year.

Within Bank of America card data, it appears the younger generations have been increasing their spending, especially on clothing, travel and restaurants, though their electronics spending has been especially strong (Exhibit 8). Some of these boosts may be temporarily driven by tax refunds (read more in Consumer Checkpoint: February bounces back), but some of the growth is likely due to previously easing constraints on consumer wallets, in our view.

Going forward, if the rise in gasoline prices seen in March persists or even intensifies, it is plausible that some of these same areas where younger generations have improved their spending could come under pressure. A key determinant, in our view, will be whether rental growth continues to cool and provides some offset.

Exhibit 8: Gen Z and Millennials have seen the most improvement in discretionary spending, especially on electronics and restaurants

Total card spending per household for select discretionary categories by household generation in February 2026 compared to February 2025 (3-month moving average, YoY%)



Source: Bank of America internal data

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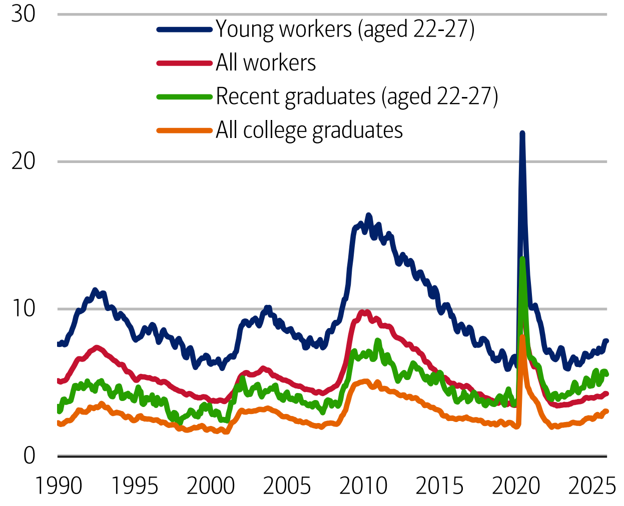
The labor market poses further challenges

Aside from gas and rent, further down the road, the labor market will likely be the most important determinant of younger generations' continued spending growth. Here there are some anxieties, especially for Gen Z.

While the overall labor market remains in a stable "low-hire, low-fire" mode, people aged 22-27 and graduating from college are currently experiencing markedly higher unemployment rates than the average worker (Exhibit 9). At the same time, sectors like retail and leisure are possibly the most exposed to a cooling in discretionary spending from a prolonged energy shock. And these sectors also tend to employ a higher share of young people (Exhibit 10). So, any slowdown due to rising fuel prices could also potentially exacerbate an already challenging labor market for the young.

Exhibit 9: The new graduate labor market is also a concern for the young

Unemployment rates by different population characteristics (monthly, %)



Source: Federal Reserve Bank of New York

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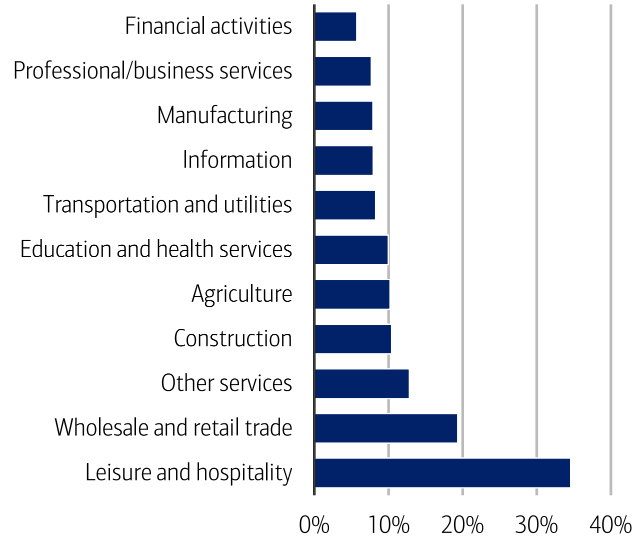
Exhibit 10: The young are a big share of consumer-facing employment

Share of employment of 16- to 24-year-olds by industry (2025, %)



Source: Bureau of Labor Statistics

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Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash, and checks.

Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under $5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

The differences between the total and per household card spending growth rate (if discussed) can be explained by the following reasons:

1. Overall total card spending growth is partially boosted by the growth in the number of active cardholders in our sample. This could be due to an increasing customer base or inactive customers using their cards more frequently.
2. Per household card spending growth only looks at households that complete at least five transactions with Bank of America cards in the month. Per household spending growth isolates impacts from a changing sample size, which could be unrelated to underlying economic momentum, and potential spending volatility from less active users.
3. Overall total card spending includes small business card spending while per household card spending does not.
4. Differences due to using processing dates (total card spending) versus transaction date (per household card spending).
5. Other differences including household formations due to young adults moving in and out of their parent's houses during COVID.

Any household consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level. Whenever median household savings and checking balances are quoted, the data is based on a fixed cohort of households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through the most current month of data shown.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards are excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

We consider a measure of services necessity spending that includes but is not limited to childcare, rent, insurance, insurance, public transportation, and tax payments. Discretionary services includes but is not limited to charitable donations, leisure travel, entertainment, and professional/consumer services. Discretionary retail includes but is not limited to general merchandise, miscellaneous, clothing, electronics, furniture. It excludes categories like groceries and gasoline. Holiday spending is defined as items in which spending in the November-December period is usually at least 20% of total annual spending on the category. Value and premium grocers are determined judgmentally on a proprietary analysis of merchants by equity analysts.

Durables spending is defined as spending on electronics, building materials, auto and furniture. Premium durables spending is based on a selection of retailers who are judged to sell relatively higher value products. Conversely, value durables spending is based on a selection of retailers who are judged to sell relatively lower value products.

For analysis looking at higher value transactions (including durables), we consider a value per transaction threshold estimated with reference to the top 30% of transactions by value in 2024. The share of higher value transactions is then the number of transactions above this threshold as a percentage of total transactions over time.

Lower, middle and higher household income cuts in Bank of America credit and debit card spending per household, and consumer deposit account data are based on quantitative estimates of each households' income. These quantitative estimates are bucketed according to terciles, with a third of households placed in each tercile periodically. The lowest tercile represents 'lower income', the middle tercile represents 'middle income' and the highest tercile 'higher income'. The income thresholds between these terciles will move over time, reflecting any number of factors that impact income, including general wage inflation, changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

Major grocery categories include sugar and sweets, juices and other non-alcoholic beverages, bakery products, processed fruits and vegetables, fresh fruit and vegetables, coffee and tea, fats and oils, milk, cereal and cereal products, other, cheese, and meats, poultry and fish, Other includes soups, snacks, frozen and freeze-dried prepared foods, and spices, seasonings, and condiments.

Generations, if discussed, are defined as follows:

1. Gen Z, born after 1995
2. Younger Millennials: born between 1989-1995
3. Older Millennials: born between 1978-1988
4. Gen Xers: born between 1965-1977
5. Baby Boomer: 1946-1964
6. Traditionalists: pre-1946

Any reference to card spending per household on gasoline includes all purchases at gasoline stations and might include purchases of non-gas items.

Additional information about the methodology used to aggregate the data is available upon request.

Contributors

Joe Wadford

Economist, Bank of America Institute

David Michael Tinsley

Senior Economist, Bank of America Institute

Sources

Li Wei

Director, Global Risk Analytics

Jon Kaplan

Senior Vice President, Consumer & Small Business Products and Analytics

Disclosures

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Economy

The business of women's sports

18 March 2026

Key takeaways

- Women's sports are evolving into a high growth segment of the broader sports economy. BofA Global Research expects more than a 250% revenue growth in US women's sports by 2030 - a trend reinforced by major sports like soccer attracting institutional and venture capital at increasing speed.
- Women's sports viewership in the US has nearly tripled since 2020. Younger, affluent and highly digital consumers are driving greater live attendance, merchandise sales, and social-media fandom, according to BofA Global Research. Women's leagues in areas like soccer, basketball, rugby and cricket are positioning the sector to grow rapidly.
- According to BofA Global Research, as viewership and engagement increase, women's sports are being met with increased sponsorship to cater to growing fanbases. And with women's discretionary spending growth outpacing men's in Bank of America credit and debit card data, this underscores women's ability to shape long-term economic opportunities.

Women's sports revenues sprinting ahead

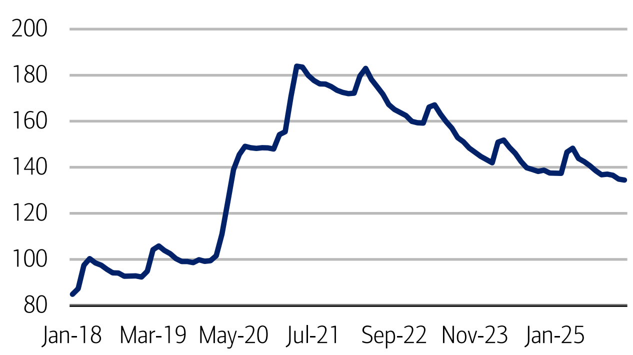
Women's sports revenues in the US could grow 250% (to $2.5 billion) between 2024 and 2030, according to a McKinsey analysis. However, as US sports add additional women's teams, as well as fans, BofA Global Research believes that a 250% increase could prove conservative. And despite this fast-paced growth, women's sports still only earn a fraction of the revenues, rights and sponsorships of men's games.

Why now? More money, more choices

Between 2018 and 2023, wealth controlled by women increased by 51%, according to a McKinsey analysis. Across the European Union and US, women currently control about one-third of retail financial assets, with this share projected to reach 40-45% by 2030.[1] Further underscoring women's financial strength, women's median deposit account balances were up 35% from the 2019 average in January 2026, per Bank of America account data. (Exhibit 1).

Exhibit 1: Women's deposit account balances were up 35% from the 2019 average in January

Women's median deposit account balance (monthly, indexed, 2019 average = 100)



Source: Bank of America internal data

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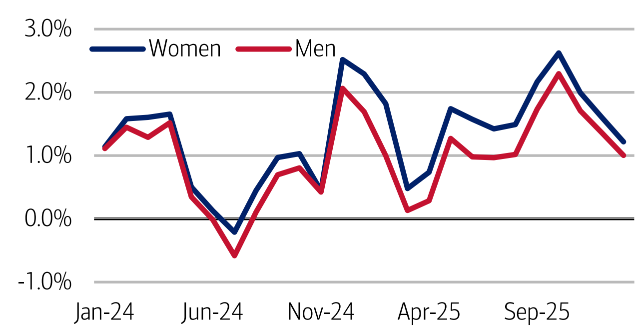
Exhibit 2: Women's discretionary spending growth was up 1.2% year-over-year (YoY) in January

Median discretionary spending by gender (monthly, 3-month moving average, YoY%)



Source: Bank of America internal data

Note: Discretionary includes all spending excluding utilities, gasoline, healthcare and groceries.

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[1] Catania, C., Zucker, J., et. al. (2025, May 8). *The new face of wealth: The rise of the female investor.* McKinsey.

Additionally, women make 70-80% of household purchases and are almost half of the women's sports fanbase, according to BofA Global Research. According to a Boston Consulting Group study, women are projected to control $32 trillion in discretionary spending globally by 2028, making them a powerful consumer segment that can transform existing sectors (read more on this in March 2025's Growing the pie). We see evidence of this in Bank of America data, as women's discretionary spending continued to outpace men's (up 1.2% YoY) in January (Exhibit 2).

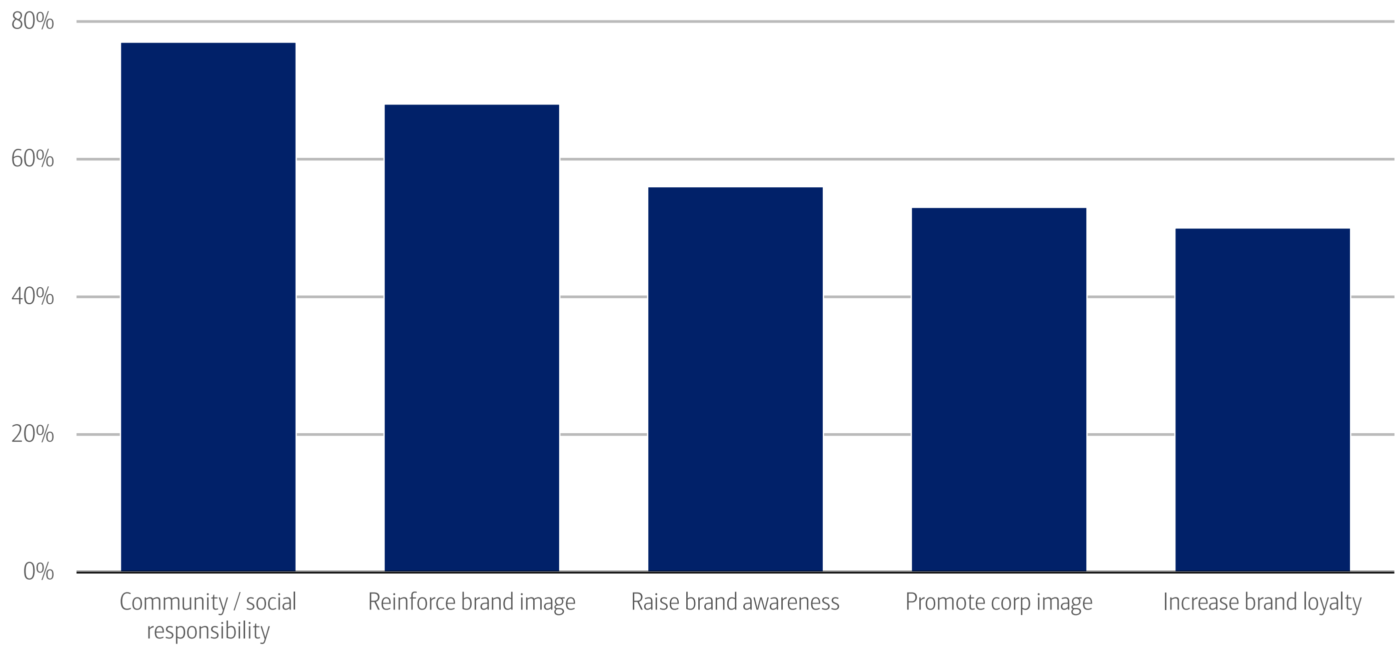
At the same time, the women's sports industry is going mainstream, with pay, sponsorships and media attention following. This is driven in part by two factors: 1) rising live-event attendance and 2) increased willingness to pay for tickets and in-person experiences (read more on the strength of in-person entertainment spending in February's On the Ball publication).

Sponsors are building out their bench

While both women's and men's professional sports make money the same way – from ticket sales, media rights and sponsorships – right now, women's sports rely most heavily on sponsorship and commercial deals.[2]

In response, brand partners are leaning into women's sports, including stakeholders from non-traditional sectors like beauty, health care, wellness and travel. One reason for this is community/social responsibility, followed closely by reinforcing brand image (Exhibit 3).

Exhibit 3: Brand decision makers highlighted social responsibility (77%), brand image (68%) and awareness (56%)
Top five reasons to sponsor women's sports



Source: Women's Sport Trust, BofA Global Research

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Additionally, brands are partnering with college athletes and innovating with collaborations across fitness, health and fashion. According to a recent Merrill study, The Financial Reality of Today's Young Athletes, 74% of respondents were expecting $25,000 or more from name, image and likeness (NIL[3]) deals, sponsorships and related opportunities in 2025. Meanwhile, 32% had already earned $10,000+ from NIL deals in 2024 – a substantial income for many young athletes (Exhibit 4).

Further sweetening the pot, a study by Change Our Game asserts that every $1 spent by a corporate sponsor generates $7 in customer value. And this adds up: the number of sponsorship deals in professional women's sports (over 5,500) rose 22% in 2024 and 12% in 2025.[4] And a 2025 report by Sports Innovation Lab states that 82% of brands who already sponsored female athletes planned to increase their spending on women's sports in the coming year.

[2] Haskel, J., Lee, P., & Giorgio, P. (2023, November 28). *Women's elite sports: Breaking the billion-dollar barrier.* Deloitte.
[3] Name, Image, and Likeness (NIL) refers to an athlete's ability to earn compensation for the use of their personal brand — specifically their name, image, and likeness — through endorsements, sponsorships, social media content, appearances and other promotional activities. Since 2021, NIL rights have allowed college and amateur athletes to monetize their public personas while maintaining eligibility, creating new financial opportunities at earlier stages of their careers.
[4] SponsorUnited. (2025, March 26). *Explosive Growth of Women's Sports Offers Bold New Opportunities for Brands.*

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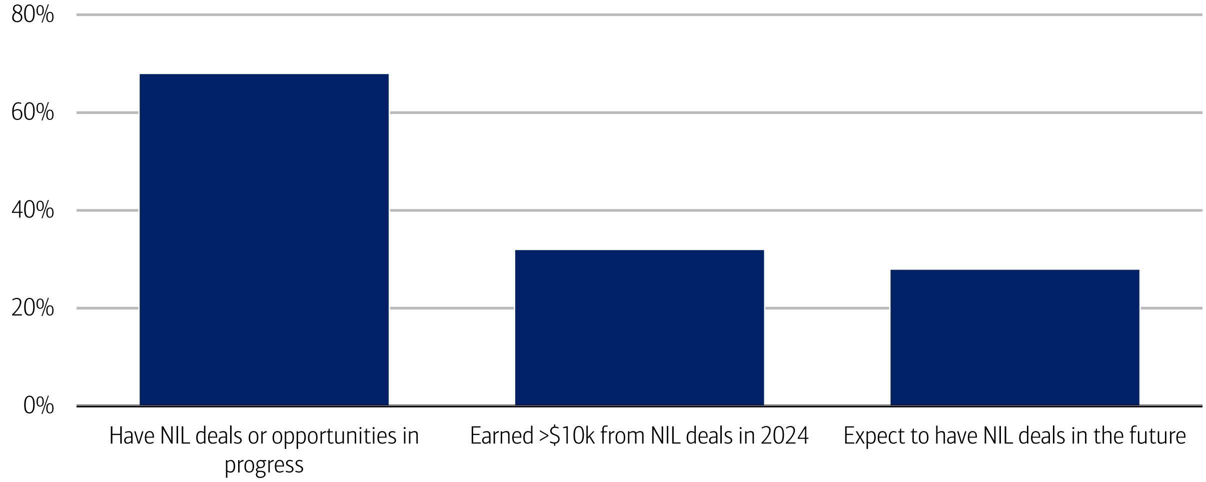
Exhibit 4: At the time of the study, 68% of respondents had NIL deals or opportunities in progress
NIL deal expectations (% of respondents)



Source: Merrill's 2025 The Financial Reality of Today's Young Athletes study
Note: See Methodology for survey details.

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Diversified global fan bases support growing market

Women's sports' popularity extends beyond just female viewers. For example, in their 2025 Global Sports Report, Nielsen noted that 53% of women's sports fans are actually men. Looking at the US, overall women's sports viewership nearly tripled from 2020 to 2025. And zooming back even further, interest in women's sports climbed to 50% of the global population in 2024, up from 45% in 2022.[5]

In general, women's sports fans tend to skew younger, more educated and more affluent compared to fans for many men's sports.[6] These viewers also tend to be tech savvy, engaging via social media and/or streaming services to watch sporting events at home (read more on streaming trends in Streaming: From trickle to torrent).

According to BofA Global Research, just 22% of female sports fans attend live events, but those who do are more engaged. They are more likely to visit fan zones, try interactive experiences, or participate in event-related activities. This can be beneficial for sponsors, as female fans are also more likely to notice branded experiences, according to The Team (formerly Wasserman) Global Sports Panel. Additionally, they are also more likely to purchase team- and player-related merchandise, and they are more likely to avidly follow (and be influenced by) their favorite players on social media.

Visibility via digital & social media: Younger generations are wide open

Today, unsurprisingly, younger people access more entertainment via digital sources than broadcast, according to BofA Global Research. And oftentimes, consumer data (e.g., engagement, watch-time, fan growth) is collected and used to implement higher fees and multi-market commitments.

The gap between old-school TV reach and what streaming can deliver is getting smaller. Big streaming platforms now pull in audiences that look a lot like broadcast TV – especially for big, one-off events, according to BofA Global Research. Plus, more fans are consuming sports via social media, and loyalty can be more player-centric than team-focused. The most common way for female fans to watch sports globally is at home with family or friends,[7] so therefore streaming platforms and digital content play a central role.

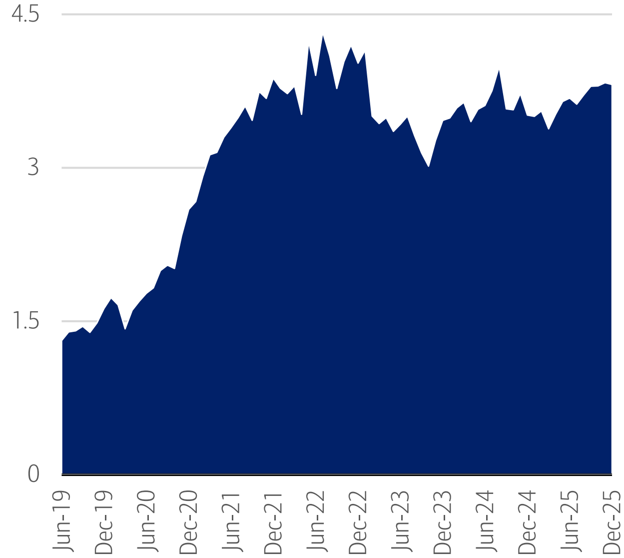
Getting in the game: Women-focused capital investment is growing

Globally, around $4 billion in equity assets under management (AUM) is now allocated to women-focused investment mandates, 2.4 times the level in 2019, according to BofA Global Research (Exhibit 5). Within impact investing, gender is an even more established theme: about 60% of impact investors, representing ~20% of their respective AUMs, allocate capital to investments supporting women and girls (Exhibit 6).

This momentum is increasingly visible in women's sports, where institutional and venture capital interest has accelerated sharply in the last couple of years, according to BofA Global Research.

[5] Nielsen. (2025, June). *The future of Sport: Nielsen's 2025 report reveals growth drivers.*
[6] The Team. (2023, August 1). *"The new economy of sports" study by The Collective Team defines the unparalleled value of pro women athletes.*
[7] The Team. (2024, October 2). *The Collective Launches Research Proving that Women are Driving the Global Growth in Sports Fandom.*

Exhibit 5: Women-focused assets under management (AUM) has grown to $4 billion (+240%) since 2019
AUM in equity-related funds globally, ($, billions, 2019-2025)



Source: EPFR, BofA Global Research

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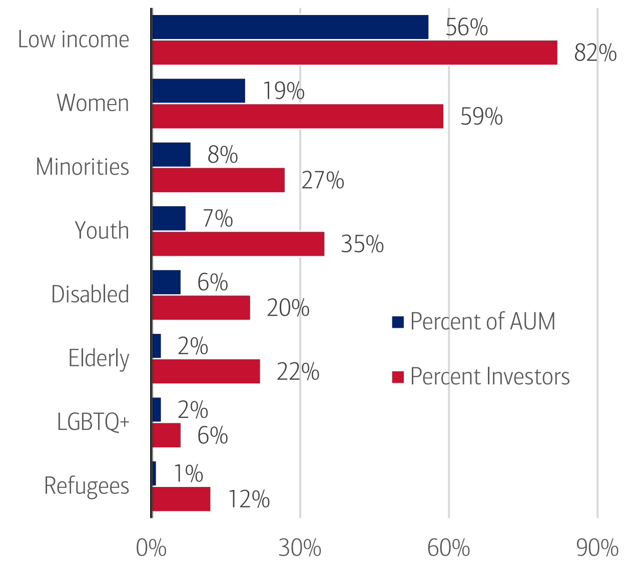
Exhibit 6: About 60% of impact investors allocate capital to investments supporting women and girls
Impact allocations by demographic group targeted (sample = 225, AUM = $90 billion, %)



Source: Global Impact Investing Network (GIIN), BofA Global Research

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Soccer strikes and scores

Women's soccer already counts over 500 million fans globally, and that figure is projected to rise to 800 million by 2030.[8] That's a 60% increase that would firmly establish women's soccer among the top five sports by fan count.[9] Interestingly, 53% of fans of women's soccer became fans within the past three years, with many having started their journey as fans of a national team.[10]

Room to grow: 20% a year through 2030?

Women's soccer has fanbase momentum, sponsor demand and early progress in media to sustain up to 20%+ annual growth through the end of the decade, according to reports from Deloitte, Visa, Nielsen-Pepsico and SportFive. Plus, women's soccer sponsorship deals jumped 42% in value from 2023 to 2024.[11]

According to BofA Global Research, women's soccer has fast-growing audiences on social media. On the sponsorship front, companies have shifted from bundled deals to more standalone or hybrid models, meaning women's sports are now being valued and contracted on their own. Unbundling increases value, brings new brands and categories, and increases overall commercial packages.[12]

Women's basketball: Fast on their feet

The WNBA (Women's National Basketball Association) has seen one of the greatest audience expansions of any US sports league ever over the past decade. Over 75 million Americans consider themselves fans, and over 60% of those fans subscribe to a streaming service specifically for sports content, according to a study by MRI-Simmons.

Regular season viewership of the WNBA more than quadrupled from 2020 to 2025 for nationally televised games; playoff games averaged 1.2 million last year. By comparison, the 2024/25 men's National Basketball Association (NBA) regular season ended with the national television networks averaging 1.53 million viewers, down 2% YoY.

Ahead of other sports, the WNBA is nationally televised across various networks and streaming services, making it one of the most accessible female sports in America. The audience is young, diverse and has a nearly even split between men and women, according to BofA Global Research.

[8] Nielsen. (2025, June). *Women's football set to enter global top 5 sports by 2030 with over 800M fans — Nielsen Sports and PepsiCo Report Reveals Untapped Opportunity for Brands.*
[9] Ibid.
[10] The Compound Effect in Women's Football. (2024, September 19). *Visa study unveils the success of women's football and its growth opportunity.* Visa.
[11] SponsorUnited. (2025, March 26). *Explosive Growth of Women's Sports Offers Bold New Opportunities for Brands.*
[12] SPORTFIVE. (2025, July 25). *Women's Football: A New Frontier for Sponsors.*

This mirrors the pattern seen in women's college basketball, which has also surged in recent years. Fans show their support by purchasing tickets, merchandise and other items, including at-home streaming subscriptions. In 2024 alone, ticket sales increased 48% and WNBA merchandise sales increased 600% from the year prior.[13]

Where do we go from here: Driving further 'womentum'

Even with the WNBA moving into the mainstream, fandom intensity remains a challenge. As unprecedented growth continues across all metrics, however, and as more teams are added to the league, there is room for sustained expansion, according to BofA Global Research.

WNBA sponsorship revenues reached $76 million in 2024, with teams averaging 44 deals each – a 52% increase since 2022.[14] Numbers are rising quickly, with league expansion just down the road: San Francisco's Golden State Valkyries were added in 2025, and five more cities will join the 13 existing teams by 2030. According to SponsorUnited, in 2024, 450 brands with 531 deals generated a deal volume increase of 17% in a single year.

Women's rugby: Going for a conversion

The 2025 Women's Rugby World Cup (RWC) in England was the biggest and most attended women's rugby tournament in history. With 444,465 tickets sold, a 330% surge in sponsorships (versus 2022 in New Zealand), record television coverage and global fan participation, the tournament redefined women's rugby's commercial and cultural standing. The sport also aims to build "big event" momentum through stronger club rivalries, marquee fixtures in major venues, and better matchday products and merchandising.

A growing fanbase that's just getting started

Almost half (49%) of all women's rugby fans joined the sport in the last two years (versus 22% for men's rugby), and 65% of women's fans have increased their engagement over the past four years.[15] While about half of women's rugby fans started as fans of the men's game, newer markets like Canada and the US are showing direct entry to the women's game.

In 2025, 15 countries televised every RWC match, and viewing hours quadrupled compared to 2021, according to BofA Global Research. Over 1.1 billion social media impressions were generated during the Rugby World Cup, and player accounts added almost 700,000 followers. Fans explicitly say player visibility increased their engagement (40% overall), especially in Australia and the US – the next two Women's Rugby World Cup hosts.

Broadcast rights accounted for 10% of 2025 World Cup revenues as the tournament reached almost 300 million households. As competitions professionalize, BofA Global Research expects that this amount has room to rise.

Cricket: Hitting it for six

Women's cricket in India is experiencing one of the fastest growth trajectories in global sports, according to BofA Global Research. The International Cricket Council's (ICC) 2025 Women's Cricket World Cup – held in India – sold roughly 300,000 tickets and had 446 million viewers for eight teams across 31 matches. In 2029, the competition will broaden with the addition of two new teams (for a total of 10). And the sport will even be featured in the 2028 Los Angeles Summer Olympics.

Sponsorship and fans are jumping

From 2025 onwards, ICC women's events have maintained separate sponsorship portfolios, reflecting confidence in the value of women's cricket.

For ICC's 2025 World Cup, sponsorship and advertising rates jumped 40-50% compared to 2022, according to BofA Global Research. Major brands across tech, consumer goods, finance, and luxury joined as top-tier partners, reflecting rising corporate interest. Cosmetics, healthcare, and lifestyle brands are also driven by rising female viewership and players' expanding personal brand value.

Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may

[13] SponsorUnited. (2025, March 4). *Women In Sports Marketing Partnerships 2024-25.*
[14] SponsorUnited. (2025, August 13). *WNBA Partnerships 2024-25.*
[15] Women's Rugby/World Rugby. (2025). *A Blueprint for Growth.*

provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Bank of America credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards is excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

If applicable, the consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level.

If applicable, any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Any reference to aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash and checks.

Median annual income growth is derived from customers who have a valid income value for every month over the time period and who have a non-null gender code. Gender data is self-select.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

Merrill's study on *The financial reality of today's young athletes* is based on research conducted among 159 high-potential athletes across the United States in 2025. High-potential athletes were defined as those age 18+ with significant earning potential, including athletes with existing or pending NIL deals of $10,000 or more, those competing at professional levels or offered professional contracts, athletes representing their country in international competition, those competing in top conferences for their sport or those expecting to pursue professional athletics. The study participants represented diverse sports including basketball (18%), football (14%), baseball (11%), soccer (9%) and various other competitive athletics. Ages ranged from 18-24, with 81% between ages 18-21. The study examined financial behaviors, NIL participation, advisor relationships and financial planning priorities, providing comprehensive insights into this unique demographic's financial needs and preferences

Additional information about the methodology used to aggregate the data is available upon request.

Contributors

Taylor Bowley

Economist, Bank of America Institute

Lynelle Huskey

Analyst, Bank of America Institute

Sources

Kay Hope

Research Analyst, BofA Global Research

Menka Bajaj

Sustainability Strategist, BofA Global Research

Haim Israel

Strategist, BofA Global Research

Dale Lin

Director, Global Risk Analytics

Adam Zeid

Vice President, Global Risk Analytics

David Plaus

Research Analyst, BofA Global Research

Disclosures

Accessible version

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Economy

Consumer Checkpoint: February bounces back

10 March 2026

Key takeaways

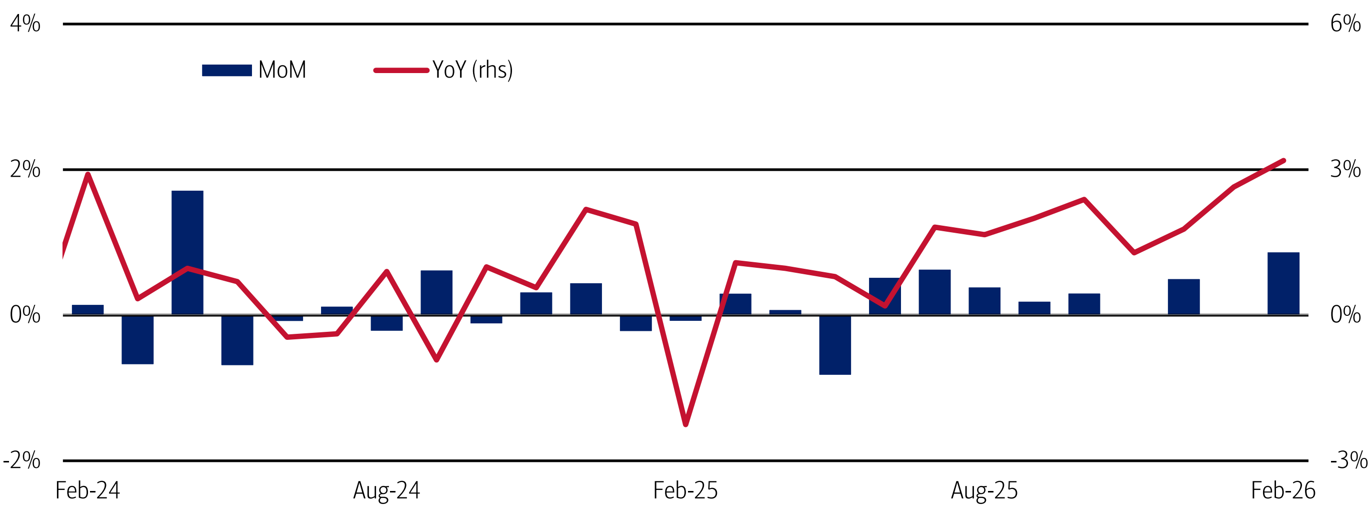
- Spending growth was very strong in February, according to Bank of America internal credit and debit card data. The year-over-year (YoY) growth rate rose to 3.2%, the highest in over three years. And on a seasonally-adjusted basis, card spending rose a strong 0.9% month-over-month (MoM).
- The "K-shape" in spending growth between higher- and lower-income households narrowed slightly in February, but remains very substantial, and a similarly large gap persists between higher- and middle-income households. In our view, divergence in wage growth underpins these trends and does not suggest that the "K" will go away any time soon.
- Average tax refunds have been larger for higher-income households so far in 2026. However, lower-income households saw larger boosts to discretionary categories than higher-income ones, likely contributing to the temporary narrowing of the "K."
- Meanwhile, most consumers still remain in good financial health in regard to their credit card capacity and savings levels, though there is a continued rise in people making minimum credit card payments, which could indicate rising stress at the margins.

Consumer Checkpoint is a regular publication from Bank of America Institute. It aims to provide a holistic and real-time estimate of US consumers' spending and their financial well-being, leveraging the depth and breadth of Bank of America proprietary data. Such data is not intended to be reflective or indicative of, and should not be relied upon as, the results of operations, financial conditions or performance of Bank of America.

Consumer momentum strengthened in February

Total credit and debit card spending per household increased 3.2% year-over-year (YoY) in February 2026 – the highest YoY growth rate since January 2023 – following the 2.6% year-over-year (YoY) rise in January of this year (Exhibit 1). Seasonally-adjusted (SA) spending per household jumped 0.9% month-over-month (MoM) in February, following a flat figure the previous month where widespread winter storms likely suppressed spending.

Exhibit 1: February's YoY spending growth was the strongest in over three years
Total credit and debit card spending growth per household, based on Bank of America internal data (monthly, MoM%, SA) and (monthly, YoY%, non-SA)



Source: Bank of America internal data

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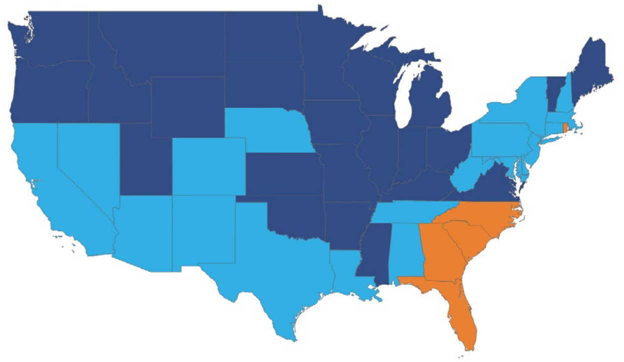
Spending held up well in the Northeast despite winter storms in late February

Overall February card-spending growth was solid across most of the US and likely benefited from a rebound in spending after January's winter weather, which affected the South and eastern seaboard. Portions of the South saw more modest growth, likely as some of the previous month's winter storm impacts appear to have spilled over into early February (Exhibit 2). However, spending was decent for most of the month in the Northeast, despite blizzards suppressing growth in New York, New Jersey, Connecticut, Massachusetts and Rhode Island toward the end of February, according to Bank of America card data (Exhibit 3).

Exhibit 2: Spending growth in the Northeast was solid YoY…
Spending growth by state (28-day moving average to February 28, YoY%, (dark blue: >4%, light blue: 4% to 2%, orange: -2% to 2%, red: <-2%))



Source: Bank of America internal data

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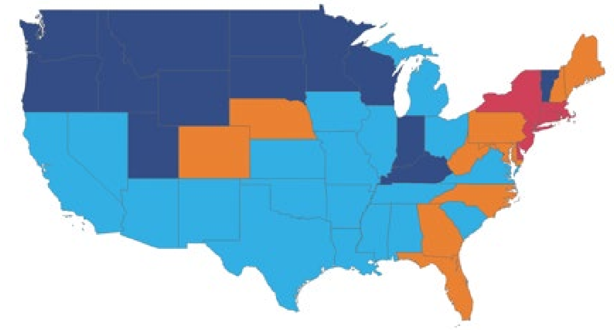
Exhibit 3: …despite notable YoY declines in late February due to winter storms
Spending growth by state (7-day moving average to February 28, YoY%, (dark blue: >4%, light blue: 4% to 2%, orange: -2% to 2%, red: <-2%))



Source: Bank of America internal data

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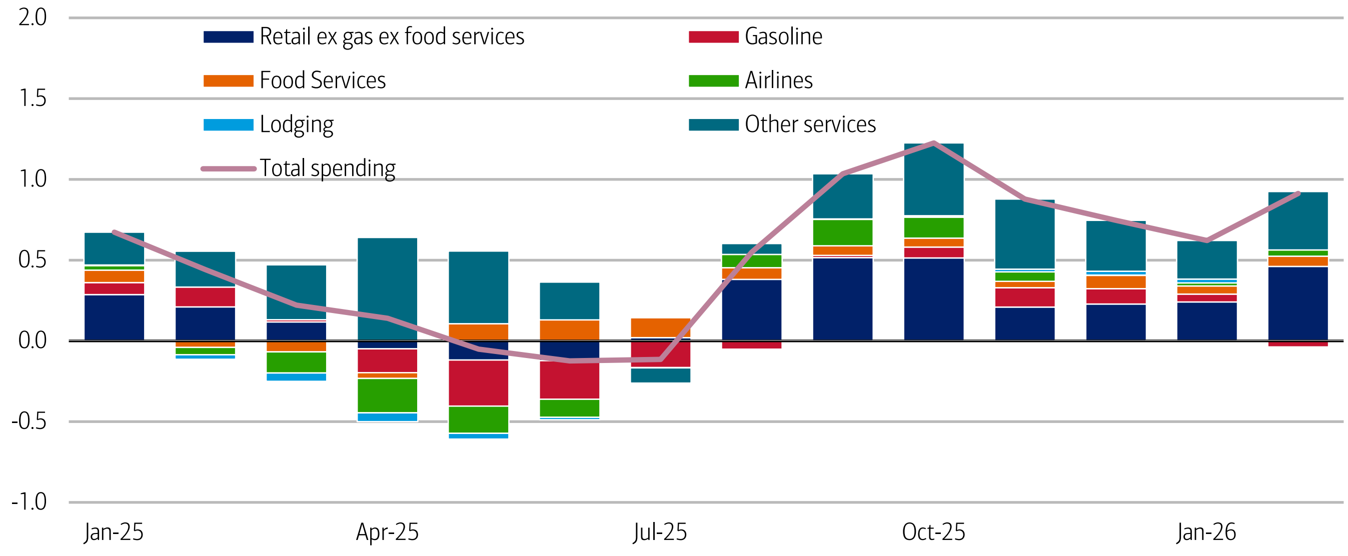
Services still in the driver's seat of card spending growth

What are people spending on? Recent momentum looks fairly balanced between services and retail spending. Services spending reflects positive contributions from highly discretionary categories like lodging, airlines and food services, and a broader range of other services (Exhibit 4), though retail spending (excluding gasoline and food services) also shows positive growth.

Gasoline spending has made little contribution to spending in recent months, up or down. But recent rises in the international price of oil and the knock-on effect on gasoline prices may put some upward pressure on household spending in this category in due course, potentially chipping away at some discretionary spending. That said, it is too early to be sure – it will ultimately depend on the extent and duration of higher prices at the pump.

Exhibit 4: Services spending is still the mainstay of overall spending momentum
Contribution to overall three-month on three-month credit and debit card spending growth by category (percentage points)



Source: Bank of America internal data

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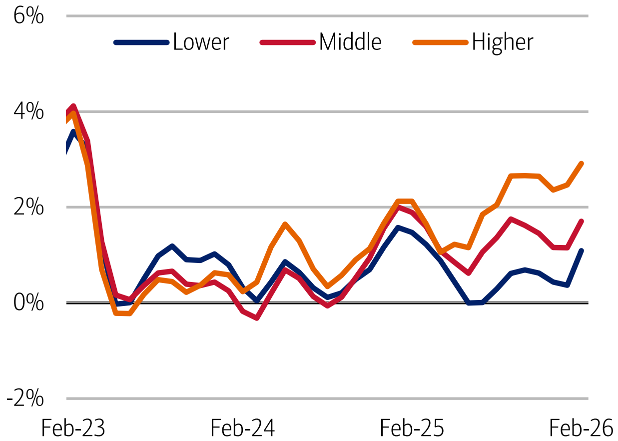
The spending "K" narrows (temporarily?)

The good news is that all income cohorts saw a rise in their three-month average YoY spending growth rates in February. In fact, the "K-shape" (or divergence) in spending growth between higher- and lower-income households narrowed slightly in February. Lower-income households spending growth was 1.1% YoY, 1.8 percentage points lower than higher-income households, at 2.9%. That is the lowest differential in growth between the cohorts in six months, but still substantial. Meanwhile middle-income households' spending growth was 1.7%, marking a 1.2 percentage point gap with higher-income households (Exhibit 5).

However, in our view, after-tax wage growth underpins these divergences in spending and suggests the underlying driver of the "K" isn't going away. In Bank of America customer deposit account data, the gap between higher-income households and all others was the largest since the data started being collected in 2015. Higher-income households' wage growth improved to 4.2% YoY in February from 3.8% YoY in January (Exhibit 6). Meanwhile, middle- and lower-income households saw wage growth cool by nearly 30 basis points to 1.2% YoY and 0.6% YoY, respectively. Notably, this is the first time that lower-income households' spending outpaced their wage growth since March 2022.

Exhibit 5: Lower-income households' spending growth accelerated to 1.1% YoY in February, a stronger improvement than the 2.9% YoY growth in higher-income households

Total credit and debit card spending per household, according to Bank of America card data, by household income terciles (3-month moving average, YoY%, SA)



Source: Bank of America internal data

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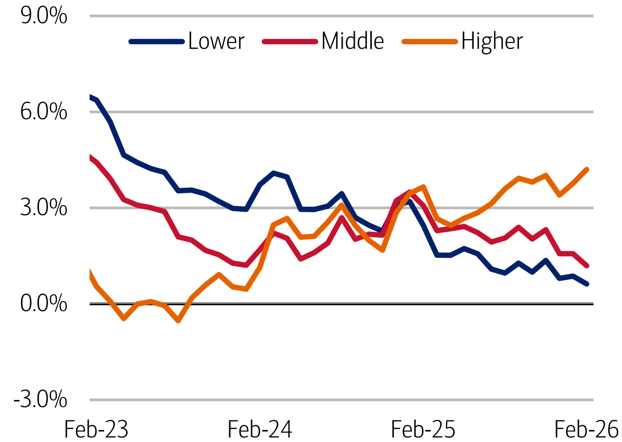
Exhibit 6: In February, lower-income households' after-tax wage growth was 0.6% YoY, while higher-income households' wage growth was 4.2% YoY

After-tax wage and salary growth by household income terciles, based on Bank of America aggregated consumer deposit account data (3-month moving average, YoY%, SA)



Source: Bank of America internal data

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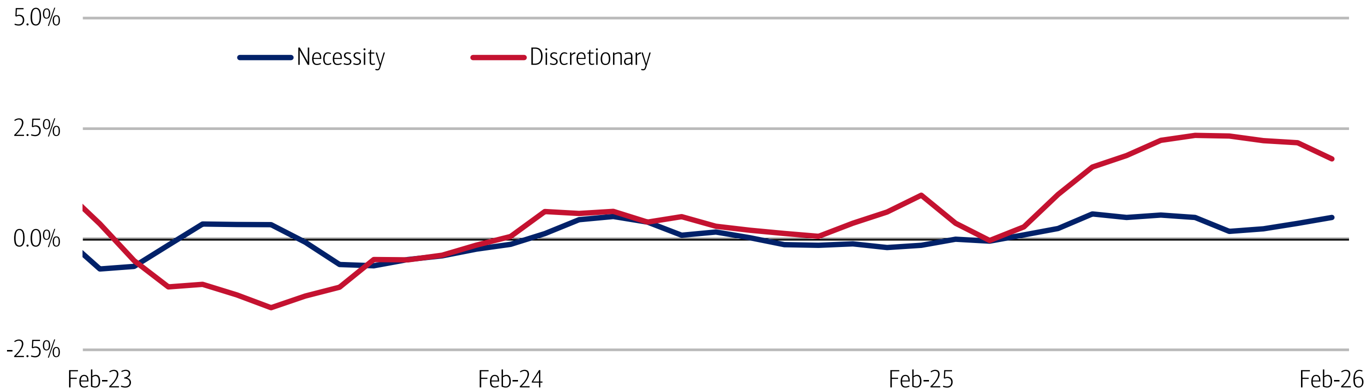
Some of the narrowing in the higher- to lower-income spending growth "K", this month, may reflect temporary boosts to lower-income households' spending after receiving tax refunds. But unsurprisingly over the past year, weaker wage growth for lower- and middle-income households is most obviously impacting their ability to spend on discretionary categories (Exhibit 7).

Exhibit 7: The "K" is kicking hardest in discretionary spending, although it has narrowed some since the beginning of the year

The difference between the card spending growth of higher- and lower-income households on necessity and discretionary categories (3-month moving average, percentage points, SA)



Source: Bank of America internal data. Necessity spending includes gas, groceries, and utilities

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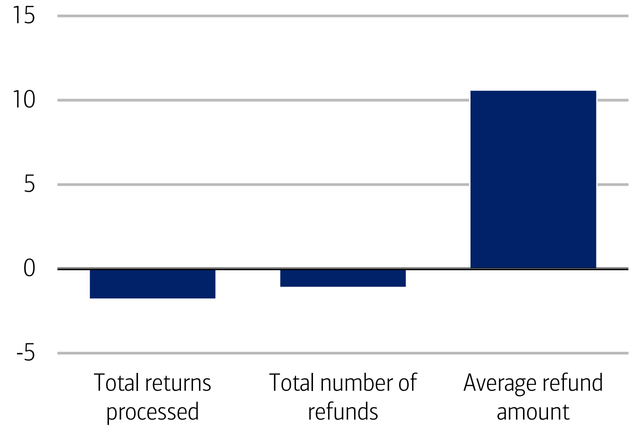
Higher-income households are seeing larger bumps to tax refunds so far

It's early in the 2026 tax season and there have been fewer tax returns processed and tax refunds issued than at the same point in 2025, according to the Internal Revenue Service (IRS) (Exhibit 8). However, so far, Bank of America deposit account data suggests that higher-income households have received larger increases in their tax refunds compared to other income cohorts (Exhibit 9). BofA Global Research also notes that the stimulus from the One Big Beautiful Bill Act (OBBBA) has so far come primarily through lower tax payments rather than refunds, a dynamic that may skew more toward higher-income households.

Exhibit 8: So far this tax season the number of payments and refunds is lagging 2025

The number of returns processed, and tax refunds issued, through February 27 compared to similar period last year (YoY%)



Source: Internal Revenue Service (IRS) data through February 27

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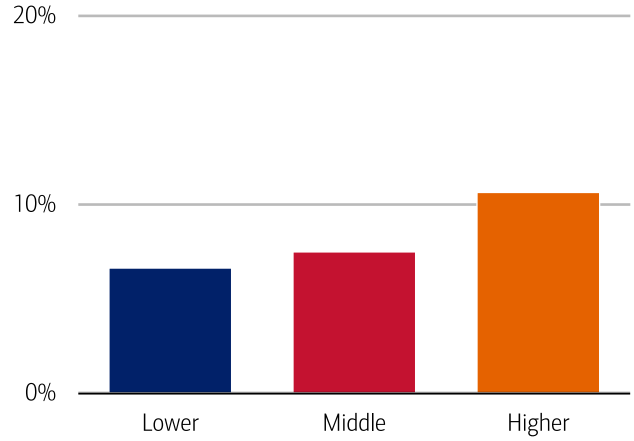
Exhibit 9: Average refunds are up most for higher-income households

Average tax refund by household income tercile through February 27 compared to similar period last year (aggregate, YoY%)



Source: Bank of America internal data

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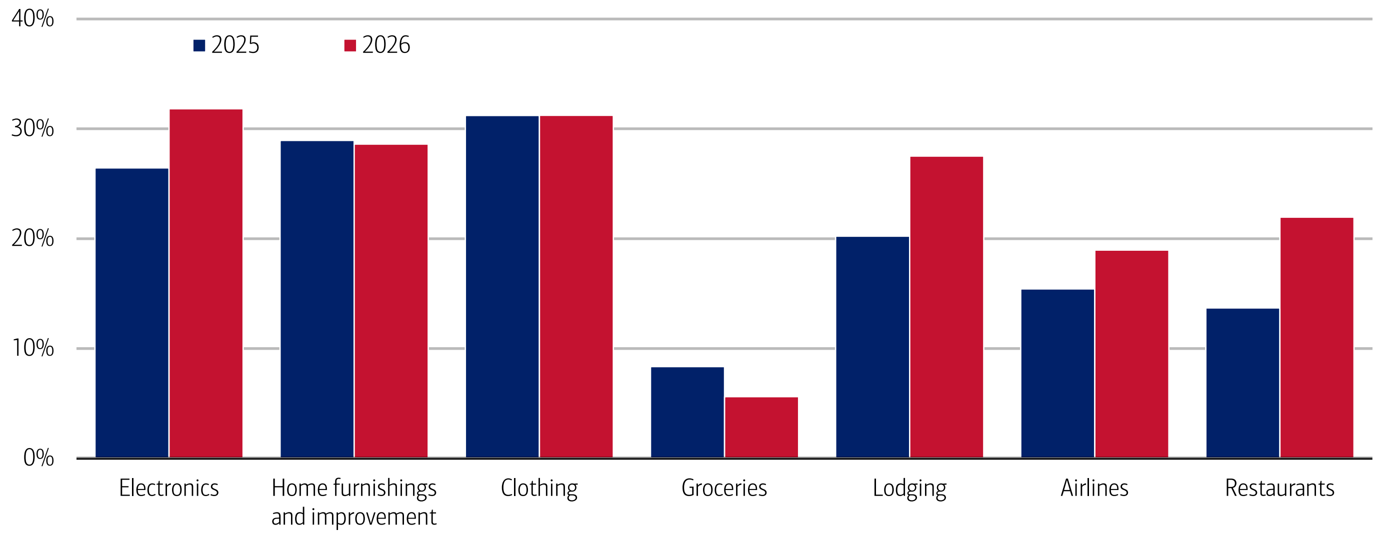
Early filers are spending on electronics and travel

Although the situation is still developing, comparing Bank of America payments data on spending before and after receipt of a tax refund gives us a sense of which spending categories get a boost from tax refunds. It appears that so far early filers have tended to favor electronics, home improvement, clothing and travel spending (Exhibit 10).

Interestingly, compared to the boost from refunds in 2025, travel and entertainment categories (lodging, airlines and restaurants) have all seen larger relative rises, with electronics seeing the largest comparative bump.

Exhibit 10: Earlier filers tend to spend slightly more on electronics, travel and restaurants

Spending across all payment channels in the three weeks after receipt of tax refund compared to average of three weeks before by select retail and service categories weekly, % change)



Source: Bank of America internal data.

Note: includes tax refunds up to February 6, 2026, compared to the entire tax period for 2025.

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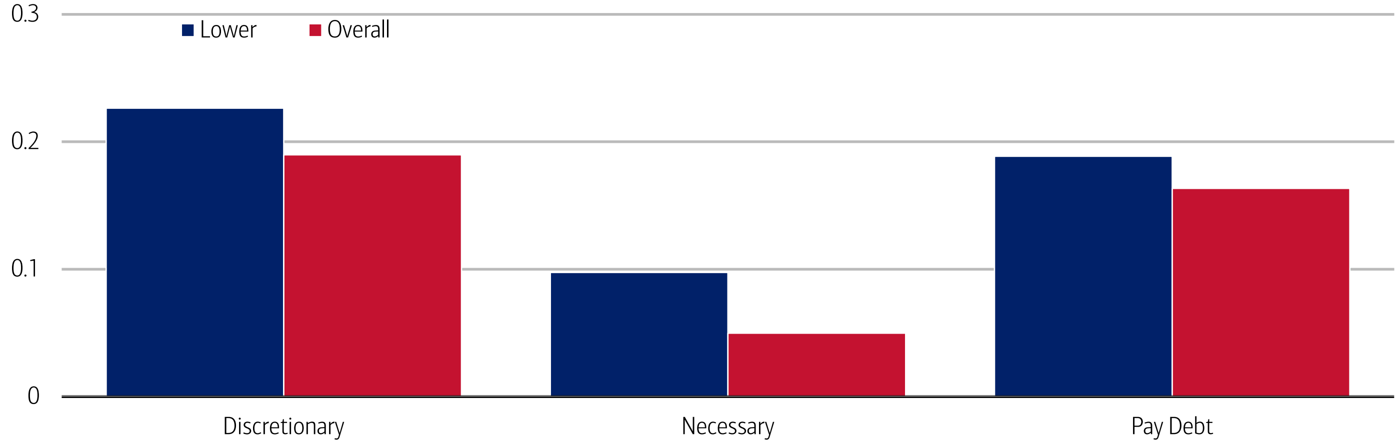
Another important note: despite average refunds rising faster for higher-income households, lower-income households have so far seen stronger boosts in spending across many discretionary categories like electronics, clothing, airlines and cruises so far in 2026 (Exhibit 11). In our view, this is likely due to tax refunds being larger relative to lower-income households' typical spending levels (read more in our publication: Consumer Checkpoint: Sloppy start, solid finish). This could also help explain some of the, likely temporary, narrowing in the overall K-shaped spending growth between higher- and lower-income households.

Exhibit 11: Lower-income households have seen a larger refund-driven spending boost relative to middle- and higher-income households
Spending across all payment channels in the three weeks after receipt of tax refund compared to average of three weeks before for select categories by household income tercile (2026, % change)



Source: Bank of America internal data
Note: includes tax refunds up to February 6, 2026. Debt payments include internal (BAC) and external debt payments.

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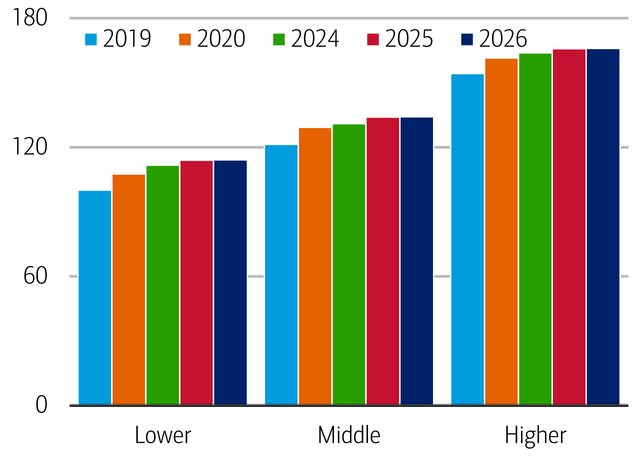
Most consumers are on top of credit card debt, with signs of stress at the margins

Post-refund debt payments have also risen across income cohorts so far, and the fact that that households are prioritizing debt repayment aligns with Bank of America internal data showing that the share of households paying their credit card balance in full each month has increased across all income levels over the past two years and relative to 2019 (Exhibit 12). This suggests to us that consumers generally maintain robust financial health. However, it is notable that the share of consumers making only minimum credit card payments has also been rising over the past two years, though this trend appears to have slowed across income cohorts (Exhibit 13).

Exhibit 12: The share of those who pay off their entire credit card balance every month is above 2019 levels for all income groups
Ratio of households who pay off their entire credit card balance every month to total households in Bank of America credit card data, by household income terciles (yearly average, index lower-income households' 2019 average = 100)



Source: Bank of America internal data
Note: 2026 data reflects January and February

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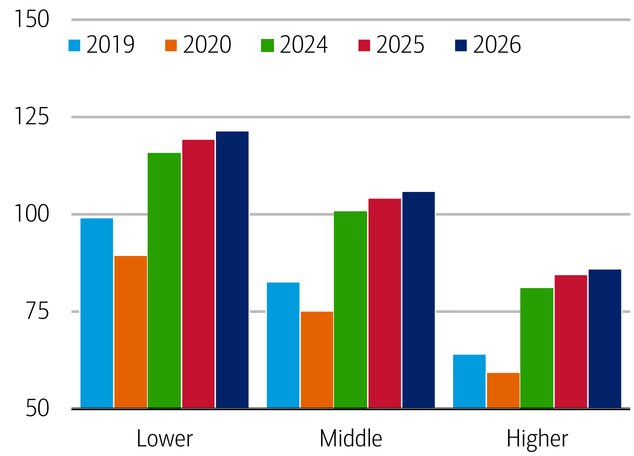
Exhibit 13: The share of minimum credit card payers is increasing in 2026*, albeit at a slower pace than last year
The share of households making minimum credit card debt payments by household income tercile (yearly, index lower-income households' 2019 average = 100)



Source: Bank of America internal data
Note: 2026 data reflects January and February

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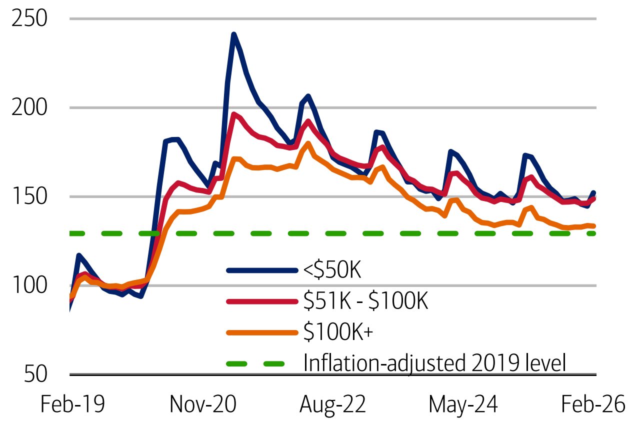
Deposit balances tend to rise over tax refund season

Typically, over the tax filing season, savings and checking balances get a boost as refunds are paid in. In Bank of America deposit data, we can begin to see signs of this in the February data. It also indicates that households across the income spectrum continue to hold large balances in both nominal and inflation-adjusted terms compared to 2019 (Exhibit 14). Notably, consumers making less than $100K a year saw the first YoY increases in their deposit levels in nearly four years, while consumers making over $100K saw only a minor decrease (Exhibit 15).

Exhibit 14: Median checking and savings deposit balances remain above inflation-adjusted 2019 levels

Monthly median household savings and checking balances by income for a fixed group of households through February 2026 (monthly, indexed 2019 = 100)



Source: Bank of America internal data

Note: Monthly data includes those households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through February 2026.

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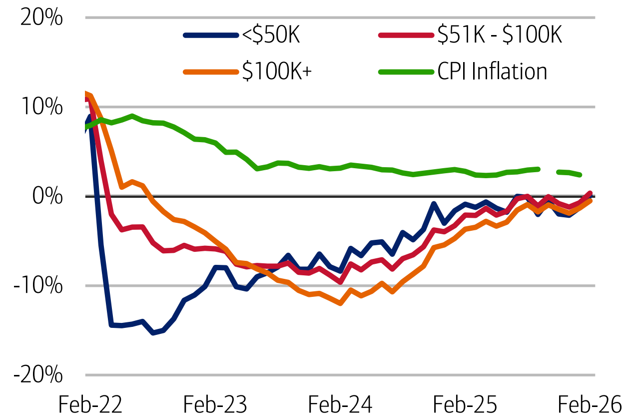
Exhibit 15: Lower- and middle-income households saw YoY increases in deposits

Monthly median household savings and checking balances by income for a fixed group of households through February 2026, based on Bank of America deposit data and Consumer Price Index inflation (CPI), based on latest Bureau of Labor Statistics data (% YoY, monthly)



Source: Bank of America internal data

Note: Monthly data includes those households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through February 2026. CPI inflation is through January.

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Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash, and checks.

Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under $5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

The differences between the total and per household card spending growth rate (if discussed) can be explained by the following reasons:

1. Overall total card spending growth is partially boosted by the growth in the number of active cardholders in our sample. This could be due to an increasing customer base or inactive customers using their cards more frequently.

2. Per household card spending growth only looks at households that complete at least five transactions with Bank of America cards in the month. Per household spending growth isolates impacts from a changing sample size, which could be unrelated to underlying economic momentum, and potential spending volatility from less active users.

3. Overall total card spending includes small business card spending while per household card spending does not.

4. Differences due to using processing dates (total card spending) versus transaction date (per household card spending).

5. Other differences including household formations due to young adults moving in and out of their parent's houses during COVID.

Any household consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level. Whenever median household savings and checking balances are quoted, the data is based on a fixed cohort of households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through the most current month of data shown.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards are excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

We consider a measure of services necessity spending that includes but is not limited to childcare, rent, insurance, insurance, public transportation, and tax payments. Discretionary services includes but is not limited to charitable donations, leisure travel, entertainment, and professional/consumer services. Discretionary retail includes but is not limited to general merchandise, miscellaneous, clothing, electronics, furniture. It excludes categories like groceries and gasoline. Holiday spending is defined as items in which spending in the November-December period is usually at least 20% of total annual spending on the category. Value and premium grocers are determined judgmentally on a proprietary analysis of merchants by equity analysts.

Durables spending is defined as spending on electronics, building materials, auto and furniture. Premium durables spending is based on a selection of retailers who are judged to sell relatively higher value products. Conversely, value durables spending is based on a selection of retailers who are judged to sell relatively lower value products.

For analysis looking at higher value transactions (including durables), we consider a value per transaction threshold estimated with reference to the top 30% of transactions by value in 2024. The share of higher value transactions is then the number of transactions above this threshold as a percentage of total transactions over time.

Lower, middle and higher household income cuts in Bank of America credit and debit card spending per household, and consumer deposit account data are based on quantitative estimates of each households' income. These quantitative estimates are bucketed according to terciles, with a third of households placed in each tercile periodically. The lowest tercile represents 'lower income', the middle tercile represents 'middle income' and the highest tercile 'higher income'. The income thresholds between these terciles will move over time, reflecting any number of factors that impact income, including general wage inflation, changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

Major grocery categories include sugar and sweets, juices and other non-alcoholic beverages, bakery products, processed fruits and vegetables, fresh fruit and vegetables, coffee and tea, fats and oils, milk, cereal and cereal products, other, cheese, and meats, poultry and fish, Other includes soups, snacks, frozen and freeze-dried prepared foods, and spices, seasonings, and condiments.

Generations, if discussed, are defined as follows:

1. Gen Z, born after 1995
2. Younger Millennials: born between 1989-1995
3. Older Millennials: born between 1978-1988
4. Gen Xers: born between 1965-1977
5. Baby Boomer: 1946-1964
6. Traditionalists: pre-1946

Any reference to card spending per household on gasoline includes all purchases at gasoline stations and might include purchases of non-gas items.

Additional information about the methodology used to aggregate the data is available upon request.

Contributors

David Michael Tinsley

Senior Economist, Bank of America Institute

Joe Wadford

Economist, Bank of America Institute

Liz Everett Krisberg

Head of Bank of America Institute

Taylor Bowley

Economist, Bank of America Institute

Vanessa Cook

Content Strategist, Bank of America Institute

Lynelle Huskey

Analyst, Bank of America Institute

Sources

Li Wei

Director, Global Risk Analytics

Yan Peng

Vice President, Global Risk Analytics

Akshita Jain

Assistant Vice President, Global Risk Analytics

Jon Kaplan

Senior Vice President, Consumer & Small Business Products and Analytics

Stephen Stancil

Senior Vice President, Consumer & Small Business Products and Analytics

Disclosures

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Economy

The Institute Employment Report: February 2026

04 March 2026

Key takeaways

- Payrolls growth accelerated to 1.3% year-over-year (YoY) in February, according to an estimate of payrolls based on Bank of America customer account data. At the same time, the growth in the number of households receiving unemployment benefits has flattened out. Overall, the impression is of a strengthening labor market in the early months of 2026.
- But there is a more concerning picture in Bank of America customer account data on after-tax wages and salaries. In particular, while higher-income wage growth rose to 4.2% YoY in February, lower- and middle-income wage growth slowed, to 0.6% and 1.2% YoY, respectively. The gap between higher-income wage growth and other cohorts is the largest it has been since the beginning of our data series.
- One reason for cooling wage growth amongst lower- and middle-income households may be weaker pay raises when changing jobs. In Bank of America internal data, the pay raise associated with a job change was 6.7% in January, down from the 2025 annual average of 8.6% and the double-digit gains during the "Great Resignation" period.

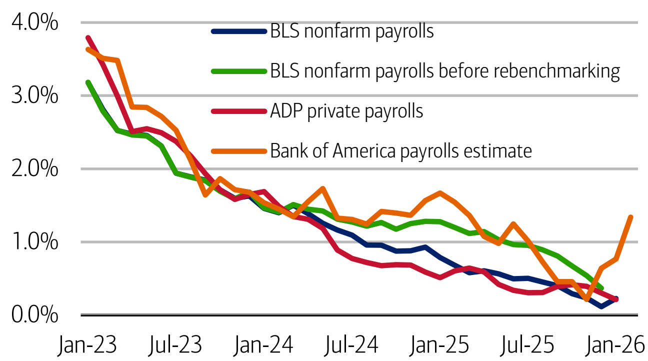
Payrolls growth accelerated again in February

Bank of America deposit account data suggests the recovery in payrolls growth we saw in January continued into February, while unemployment payments growth remained relatively flat. But the picture on wage growth in our data was less encouraging.

We use Bank of America consumer deposit data to estimate a payrolls series by looking at how the number of customer accounts receiving a paycheck is changing (see methodology). This data can be fairly noisy, partly due to seasonal variation. However, looking at a three-month moving average, Exhibit 1 suggests that the year-over-year (YoY) growth in our measure rose to 1.3% in February 2026, up from the 0.8% YoY in January.

Exhibit 1: Bank of America internal data continues to suggest an ongoing job growth rebound

Payroll estimates from Bank of America internal data (three-month moving average, % YoY), the Bureau of Labor Statistics (BLS) and Automatic Data Processing (ADP) (monthly, YoY)



Source: Bank of America internal data, Haver Analytics

Note: BLS and ADP data are seasonally adjusted, Bank of America data is not seasonally adjusted.

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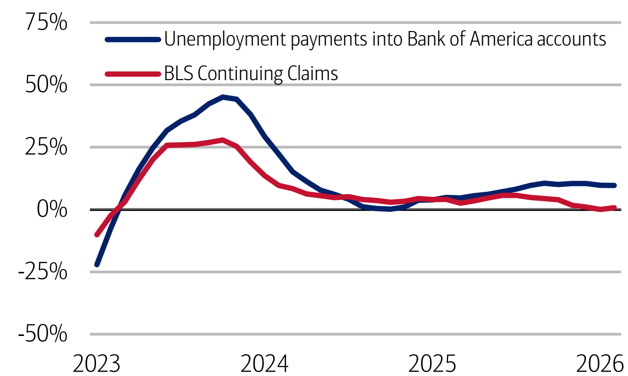
Exhibit 2: Unemployment payments into Bank of America customer accounts show fairly steady growth

Number of households receiving unemployment payments (three-month moving average, YoY%, not seasonally adjusted (NSA)) and Continuing Claims (three-month moving average, YoY%, seasonally adjusted (SA))



Source: Bank of America internal data, Bloomberg

Note: February Continuing Claims YoY data is average for weeks through February 13, 2026.

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The latest Bureau of Labor Statistics' (BLS) payroll estimate was revised lower following the benchmark revision process. This downward adjustment means BLS payrolls growth is now estimated to be lower in 2024/25 than previously estimated, implying our estimate of payrolls growth appears stronger than the official data over the period. Nonetheless, in our view, the positive growth in our estimate is a useful directional signal, indicating that the official data may have upside growth potential in coming releases.

Bank of America data on unemployment payments into customer accounts is broadly consistent with an improvement in the labor market. Exhibit 2 shows the growth in unemployment payments into Bank of America customer accounts has leveled-off at just below 10% YoY.

But the slowdown in middle-income households' wage growth has quickened

While the picture on stronger payrolls growth is "good news," there is a concerning picture coming from Bank of America deposit data on households' after-tax wage and salary growth.

One concern is that the gap between higher- and lower-income households' after-tax wage growth is wide. In fact, in the February data, higher-income after-tax wage growth accelerated to 4.2% YoY, while lower-income after-tax wage growth dropped back to 0.6% YoY. This means the gap between higher- and lower-income households wage growth was 3.6 percentage points (pp) – the highest of any point in our data, which begins in 2015.

As we discussed in last month's Employment Report, signs that middle-income households' after-tax wage growth was softening were also a concern. And in the February data we have seen this trend accelerate. Middle-income after-tax wage growth fell to 1.2% YoY in February, making the gap with higher-income wage growth also the widest it has been over the length of our series.

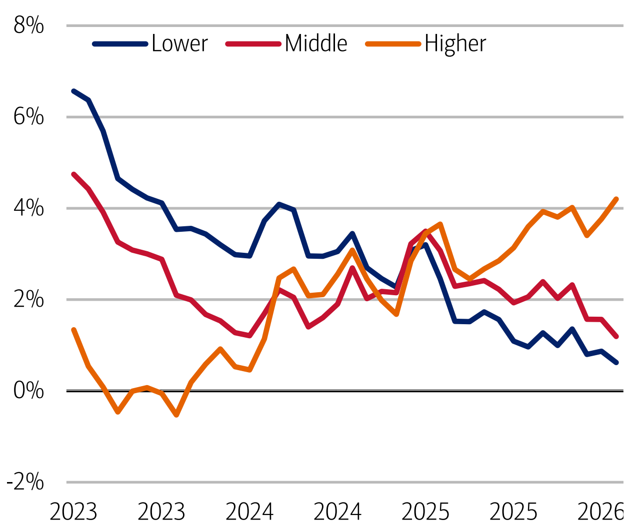
What's driving softer after-tax wage growth for lower- and middle-income households in our data? In part, it appears to be smaller raises for those changing jobs. In Bank of America deposit data, the pay raise associated with a job change was 6.7% in January, down from the 2025 annual average of 8.6% and the double-digit gains during the "Great Resignation" period in 2021-22 (Exhibit 4).

It could be that these softer gains from switching jobs are landing hardest on lower- and middle-income households, given these groups often achieve wage progression through job moves. We also discuss job-to-job pay raises and the differences between men and women in our publication: Where are women positioned in a "K-shaped" economy?.

While the short-run outlook for consumers is being buoyed by higher tax refunds, how the dichotomy in our data between strong jobs growth and growing income divergence develops will, in our view, be key later this year.

Exhibit 3: In February, lower-income households' after-tax wage growth was 0.6% YoY, while higher-income households' wage growth was 4.2% YoY

After-tax wage and salary growth by household income terciles, based on Bank of America aggregated consumer deposit account data (3-month moving average, YoY%, SA)



Source: Bank of America internal data

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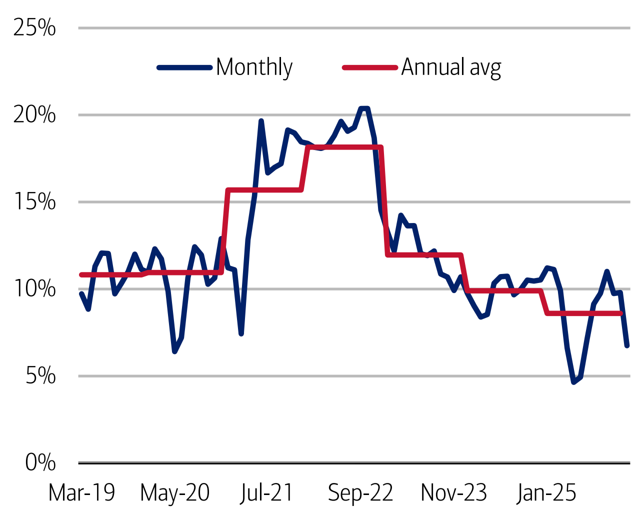
Exhibit 4: In 2025, the annual average pay raise associated with a job change was 8.6%, and in January 2026, this dropped further to 6.7%

Median pay raise for job-to-job movers (%, monthly, 3-month moving average, and annual average)*



Source: Bank of America internal data

*Calculated as the change in pay in the three months from a job move compared to pay over the same three months a year earlier.

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Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash, and checks.

Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under $5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

The differences between the total and per household card spending growth rate (if discussed) can be explained by the following reasons:

1. Overall total card spending growth is partially boosted by the growth in the number of active cardholders in our sample. This could be due to an increasing customer base or inactive customers using their cards more frequently.

2. Per household card spending growth only looks at households that complete at least five transactions with Bank of America cards in the month. Per household spending growth isolates impacts from a changing sample size, which could be unrelated to underlying economic momentum, and potential spending volatility from less active users.

3. Overall total card spending includes small business card spending while per household card spending does not.

4. Differences due to using processing dates (total card spending) versus transaction date (per household card spending).

5. Other differences including household formations due to young adults moving in and out of their parent's houses during COVID.

Any household consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level. Whenever median household savings and checking balances are quoted, the data is based on a fixed cohort of households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through the most current month of data shown.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards are excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

We consider a measure of services necessity spending that includes but is not limited to childcare, rent, insurance, public transportation, and tax payments. Discretionary services includes but is not limited to charitable donations, leisure travel, entertainment, and professional/consumer services. Holiday spending is defined as items in which spending in the November-December period is usually at least 20% of total annual spending on the category.

For analysis looking at higher value transactions (including durables), we consider a value per transaction threshold estimated with reference to the top 30% of transactions by value in 2024. The share of higher value transactions is then the number of transactions above this threshold as a percentage of total transactions over time.

Lower, middle and higher household income cuts in Bank of America credit and debit card spending per household, and consumer deposit account data are based on quantitative estimates of each households' income. These quantitative estimates are bucketed according to terciles, with a third of households placed in each tercile periodically. The lowest tercile represents 'lower income', the middle tercile represents 'middle income' and the highest tercile 'higher income'. The income thresholds between these terciles will move over time, reflecting any number of factors that impact income, including general wage inflation, changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

Generations, if discussed, are defined as follows:

1. Gen Z, born after 1995
2. Younger Millennials: born between 1989-1995
3. Older Millennials: born between 1978-1988
4. Gen Xers: born between 1965-1977
5. Baby Boomer: 1946-1964
6. Traditionalists: pre-1946

Any reference to card spending per household on gasoline includes all purchases at gasoline stations and might include purchases of non-gas items.

Estimate of payrolls growth from Bank of America internal data is based on the change in customer accounts receiving a paycheck in the month. An adjustment is made for the difference between overall population growth and customer account growth.

The job-to-job change rate (j2j rate) is defined as the proportion of customers with an identified change in their employer as a proportion of the total number of customers with employment income. We estimate the median pay rise associated with a j2j change using the pay in the first three months of the new job compared to the same three months a year ago.

Additional information about the methodology used to aggregate the data is available upon request.

Contributors

David Michael Tinsley

Senior Economist, Bank of America Institute

Taylor Bowley

Economist, Bank of America Institute

Sources

Li Wei

Director, Global Risk Analytics

Yan Peng

Vice President, Global Risk Analytics

Akshita Jain

Assistant Vice President, Global Risk Analytics

Jasmine Elder

Vice President, Global Risk Analytics

Dale Lin

Director, Global Risk Analytics

Adam Zeid

Assistant Vice President, Global Risk Analytics

Disclosures

Economy

Where are women positioned in a “K-shaped” economy?

03 March 2026

Key takeaways

- Women drove much of the labor force expansion in 2025, as hiring persisted in female-dominated sectors like private education and healthcare. This places women at the center of current labor market strength.
- The slowdown in job switching has sharply reduced the ability of workers to secure higher wages, according to Bank of America deposit account data. In January, the pay increase from a job-to-job change rate was less than half the 2019 average, and, while there is evidence of the gender pay gap narrowing, it's in part because overall wage momentum has weakened.
- As affordability pressures rise and income growth moderates, women are leaning more heavily toward "value" spending. Women's shift toward value has been strongest in apparel, reflecting how labor market pressures are spilling into household budgets.

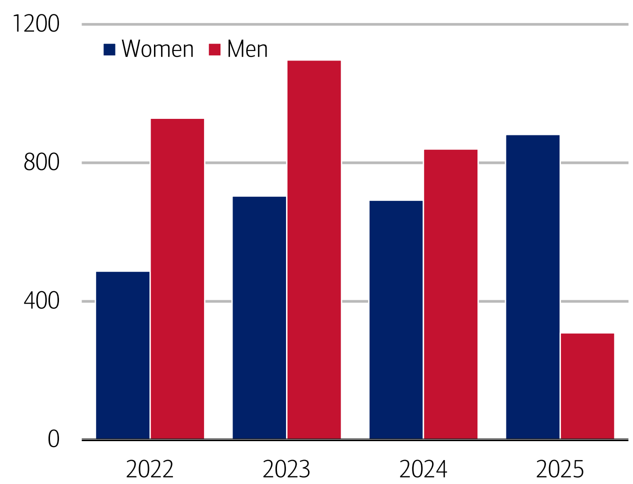
Job gains are concentrated in female-dominated sectors

Women are once again supporting labor market fundamentals. In 2025, the number of jobs added from women was nearly three times the rate of men, according to data from the Bureau of Labor Statistics (BLS) (Exhibit 1). This reversed a trend from over the past three years. In fact, from January 2025 to January 2026, women saw gains in all industries but one where they comprise more than 50% of employment (the exception was financial).

What’s driving this? Notably, big gains in private education and healthcare employment (Exhibit 2), a sector in which women comprise 77% of employment as a share of all jobs. In fact, job growth in health care averaged 33,000 per month in 2025 – more than double that of overall payrolls, according to the BLS.

Exhibit 1: For the first time in three years, women joined the labor market in 2025 at nearly three times the rate of men

Civilian labor force aged 16-64 number of jobs added from January-December by sex (thousands, seasonally adjusted, annual)



Source: Haver Analytics

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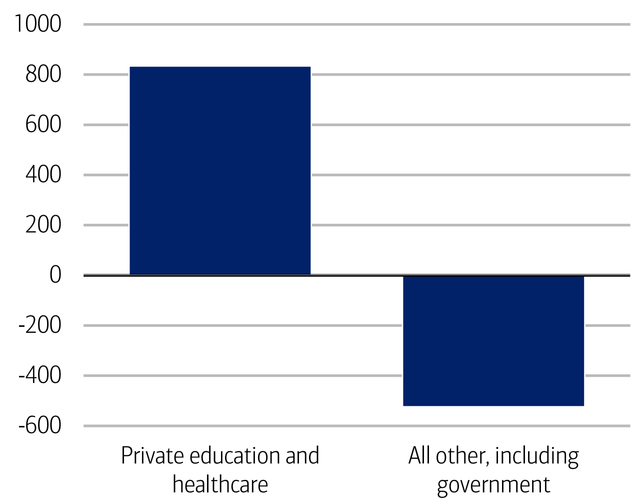
Exhibit 2: Private education and healthcare have contributed to the strongest job gains for more than the past year

Change in employment by sector from January 2025-January 2026 (seasonally adjusted, thousands)



Source: Haver Analytics

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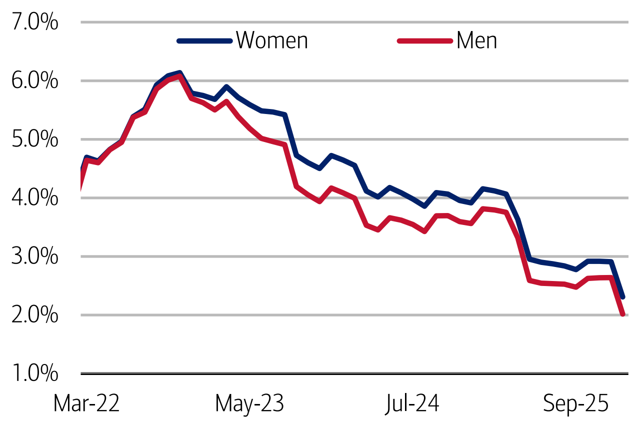
A narrowing in the gender pay gap: A double-edged sword

An increasingly important lens on today's cooling labor market is how pay growth is evolving. For both men and women, the rate of after-tax pay increases – whether tied to staying in a job or switching roles – has moderated meaningfully compared to last year (Exhibit 3).

So too has the rate at which people are changing jobs (read more on this in [August's publication on Job hoppers](#)). But there have been small signs of reacceleration in the labor market as of late, and, in January, the job change rates (see Methodology) for both men and women were above the 2025 average and in line with the 2022 average, supporting evidence of possible renewed momentum (Exhibit 4).

Exhibit 3: For both women and men, rate of pay increases have moderated in the past two months after holding steady in the second half of 2025

After-tax pay change by gender (monthly, three-month moving average, %)



Source: Bank of America internal data

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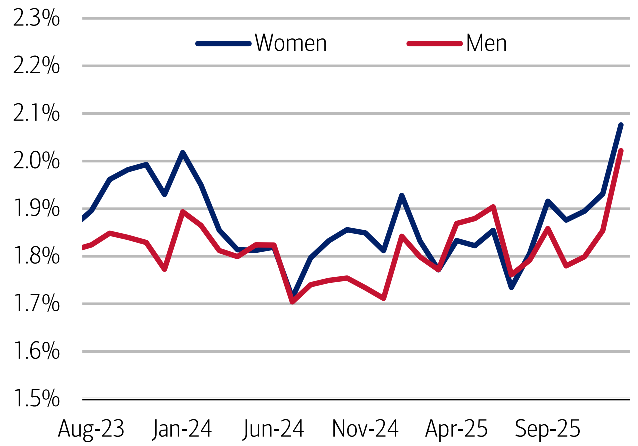
Exhibit 4: In January, the job change rate for women was 2.1% and 2.0% for men

Job change rate by gender (monthly, six-month moving average, %)



Source: Bank of America internal data

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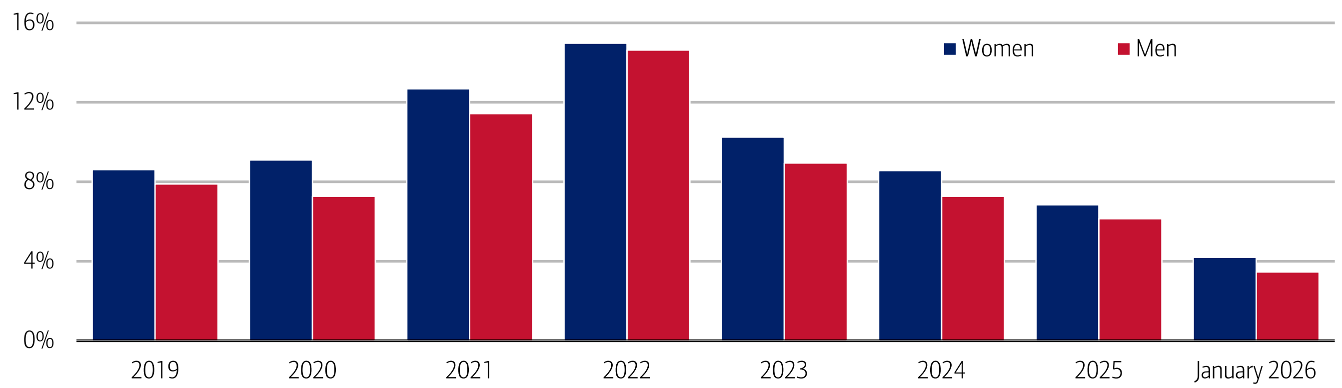
A tougher job market makes pay gains harder to come by

The slowdown in wage increases and job-change activity reflect a labor market that has continued to normalize after several years of momentum. With fewer open roles, the job-change premium – the extra pay boost workers typically receive when they switch jobs – has started to compress across the board. This softening matters because job changing remains one of the most effective ways workers secure higher pay.

Men have recently seen a slightly stronger increase in the year-over-year growth in job-change rates, giving them somewhat more bargaining power to tap into the still-remaining pockets of opportunity for higher pay. Even so, according to Bank of America deposit account data, the median pay raise associated with a job change has moderated for both men and women, with January's level measuring less than half that of the 2019 average (Exhibit 5).

Exhibit 5: For both men and women, the pay raise associated with a job change continues to moderate

Median pay raise for job-to-job movers by gender (%, annual average)*



Source: Bank of America internal data

*Calculated as the change in pay in the three months from a job move compared to pay over the same three months a year earlier.

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Women maintaining a modest edge in job-change pay raises is a positive sign for narrowing the gender pay gap, though it may also reflect differences in starting salary levels or sector mix, both of which influence percentage gains (read more on this in [March's Women and wealth](#) publication).

Looking ahead, if "low-hire, low-fire" continues to characterize the labor market, the job-change premium could compress further, limiting the extent to which workers can secure meaningful pay bumps by switching roles.

Unemployment holds steady, but the gender gap in pay disruptions has widened

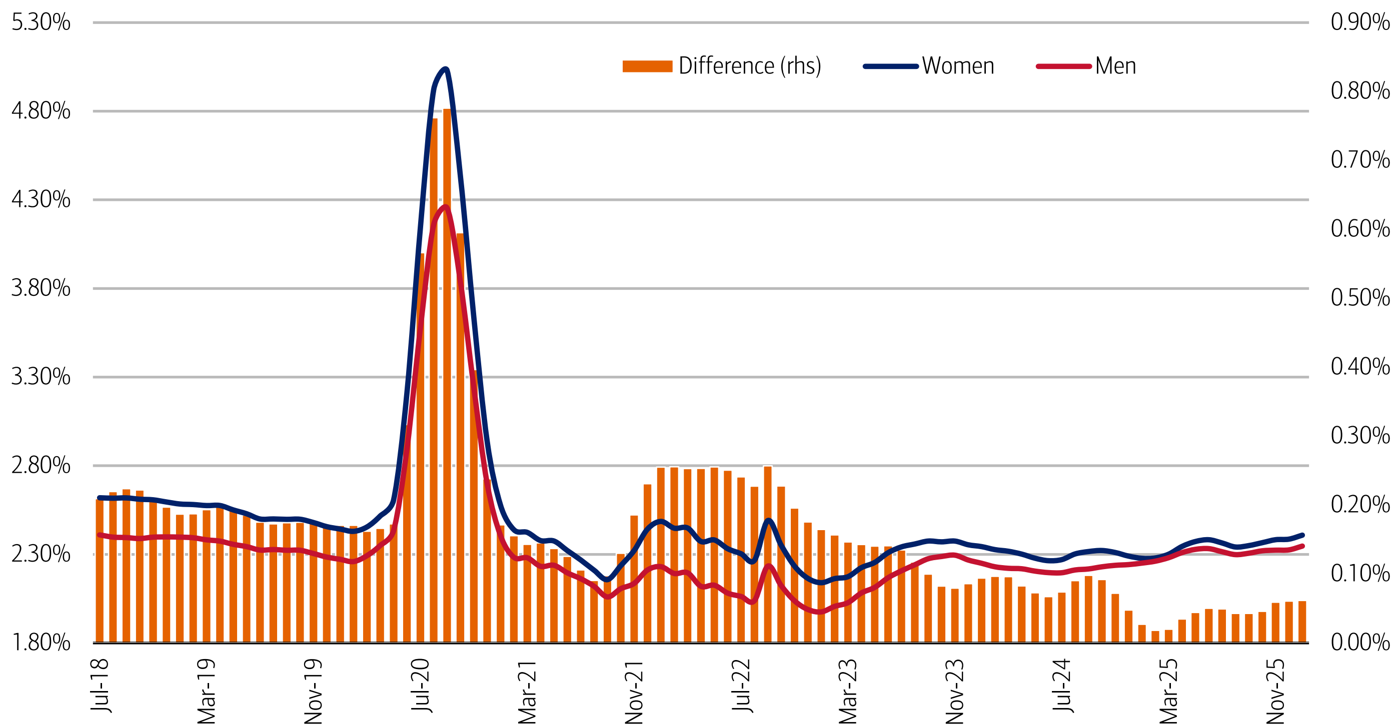
Although the overall unemployment rate has not set off any alarm bells, with growing evidence of a divergence in labor market trends across income cohorts, we also see a difference in participation between men and women in the workforce.

After having narrowed in 2025, the gap between women and men's pay disruption rate which, in our view, acts as a proxy for someone who has either temporarily or permanently left the labor force (see Methodology - Pay Disruption Rate), has widened marginally and both rates have increased over the past year (Exhibit 6).

Those increases mean that people are now more likely to experience a pause in pay – a clear marker of a softening labor market. It may also, perhaps, be an early sign of a slightly uneven labor market between men and women. Still, the good news is for both the pay disruption rate and difference between them remains below 2019 levels, according to Bank of America deposit account data.

Exhibit 6: After having narrowed in 2025, the gap between women and men's pay disruption rate has widened marginally in January

Pay disruption rate by gender (seasonally adjusted three-month moving average, %, monthly and difference (right-hand side (rhs), monthly, three-month moving average, %))



Source: Bank of America internal data

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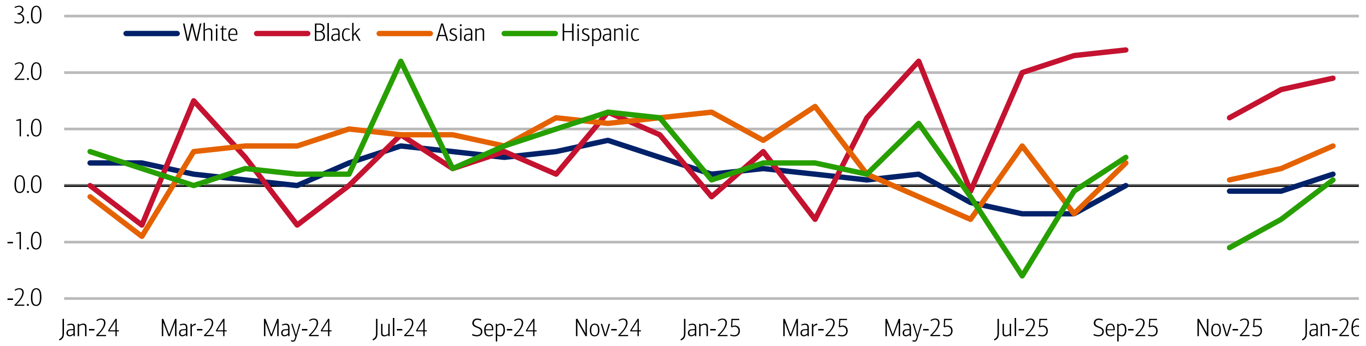
Digging deeper: Black women's unemployment rate rising fastest

It's important to note that among women, there are differences in the labor market. Over the past year, prime-age (25-54) Black women have experienced greater employment losses than other groups of women, according to data from the BLS (Exhibit 7).

While Black women saw a net increase in private-sector employment in 2025 – primarily in the growing education and health services industry – there were net losses in six of the 12 major private-sector industries, according to an analysis by the Economic Policy Institute, a nonpartisan think tank.

About one-third of the losses were in other services, a category which includes jobs in beauty and personal services, followed by 25% in manufacturing, 21% in financial activities, and 15% in professional and business services.[1] The public sector was also a notable contributor to these losses.

Exhibit 7: Black women's growth in the unemployment rate has outpaced other women by a wider margin since June 2025
Unemployment rate of prime-age women by race (not seasonally adjusted, percent change from a year ago, monthly, %)



Source: Haver Analytics
Note: The gap in data is due to the government shutdown in October 2025.

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Women prioritize stretching their dollars

Stronger income growth for women – either through job changes or otherwise – drives further financial stability and fuels spending growth (read more on this in September's A window into women's pay and purchasing power report). And weaker labor market conditions weigh on spending.

With affordability pressures on the rise (read more on this in February's Consumer Checkpoint), trading down remains a key strategy when it comes to spending, and we see evidence of that in a proprietary gauge that ranks spending across "premium," "standard" or "value" tiers, which we call the spending tier composite (STC, see Methodology).

Based on our analysis, "value" continues to gain appeal overall, but the gap between men and women's STC has widened (Exhibit 8). And though the difference between STC by gender has come down since pre-pandemic, it is starting to marginally increase once again, with women still seeking value when they shop slightly more than men.

According to Bank of America data, across all categories except apparel, women's STC has improved over the past few months, with grocery having the strongest gains (Exhibit 9). These patterns reinforce the narrative that as labor market momentum slows, the trend of consumers seeking value is likely to persist.

Exhibit 8: The difference between men and women's STC has continued to widen since 2021, suggesting women seek value more often than men
Difference between men and women's STC (monthly, 3-month moving average)

Source: Bank of America internal data

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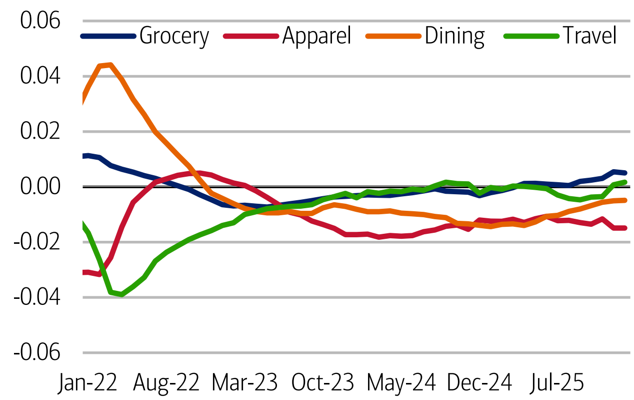
Exhibit 9: Across all categories except apparel, women's STC has improved over the past few months
Women's STC by spending category (percent difference from prior year, monthly, 3-month moving average)



Source: Bank of America internal data

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[1] Wilson, V. (2026, February 10). *Black women suffered large employment losses in 2025—particularly among college graduates and public-sector workers* | Economic Policy Institute

Methodology

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Three tiers (premium, standard and value) were based on after-tax median income derived from payroll direct deposit of individual customers who have shopped at such stores. The stores were then ranked by the median income of their shoppers, with the top third denoted as "premium," the middle third as "standard," and the bottom third as "value." In our view, such categorization is a fair view of how expensive the items are at those stores. Any stores included has had at least 100,000 individual Bank of America customers making at least one purchase during the past 12 months.

The sample of customers in this analysis includes a dynamic pool of customers that have a checking, a saving or a credit card account with BAC each month. Each customer's tier was determined by taking customer spending during the past 12 months, across the three tiers. The tier with the highest percent of spending will determine the customer's tier for each category. For example, if the customer spent the majority of their apparel dollars at premium tier apparel stores, the customer's apparel tier is designated as premium tier, even though they might still have apparel spending at the value/standard tier. This is repeated for dining (restaurants, bars, etc.), travel (hotel lines, car rental agencies, airlines, etc.), and grocery stores.

For the "spending tier composite", or STC, if a person spends most of their money at the premium tier, that equals a score of "3," while standard is a "2" and value equals "1." This is done across four major spending categories: groceries, apparel, restaurants and/or travel. The highest score that can be achieved is a "12" (if someone spends the majority of their money at premium stores across all four categories), while the lowest is a "3" (if another person spends the majority of their money at value stores for groceries, restaurants, and apparel and has no travel spending). Then, we take an average of everyone in our sample to get a spending tier composite.

If applicable, the consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level.

If applicable, any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Any reference to aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash and checks.

Median annual income growth is derived from customers who have a valid income value for every month over the time period and who have a non-null gender code. Gender data is self-select.

The Pay Disruption Rate is defined as the proportion of customers who previously had 12 months of regular payroll payments into their accounts, followed by at least 3 and up to 6 months of no payments, relative to the total number of customers with 12 consecutive months of payroll.

The job-to-job change rate (j2j rate) is defined as the proportion of customers with an identified change in their employer as a proportion of the total number of customers with employment income. We estimate the median pay rise associated with a j2j change using the pay in the first three months of the new job compared to the same three months a year ago who have a non-null gender code.

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Contributors

Taylor Bowley

Economist, Bank of America Institute

Sources

Dale Lin

Director, Global Risk Analytics

Adam Zeid

Assistant Vice President, Global Risk Analytics

Disclosures

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Transformation

Physical AI, part 3: The future of mobility

26 March 2026

Key takeaways

- Physical AI is transforming mobility, shifting vehicles from software-defined machines to AI-defined platforms with reusable intelligence that spans autos, trucks, drones and industrial equipment. This marks a structural transition where anything that moves is becoming autonomous.
- Robotaxis, autonomous trucks and drone networks are accelerating toward commercialization, supported by falling hardware costs, maturing regulation and scalable software infrastructure. These advancements are reshaping value chains, enabling new business models and expanding multi-billion-dollar addressable markets.
- In this third and final part of our physical AI series, we spotlight key trends that are gaining rapid commercial momentum and redefining the future of mobility.

Physical AI and the future of mobility

Today, physical AI is reshaping mobility, from AI-defined vehicles to highly automated robotaxis and fully autonomous fleets. Those same capabilities are rapidly spreading to trucks, delivery robots and a growing class of machines, signaling a structural shift: anything that moves is going autonomous.

Vehicles represent one of the first scaled deployment environments for physical AI – where perception, world modeling, planning and safety-critical execution converge. While fully autonomous vehicle (AV) technologies are beginning to scale, higher volume deployments of physical AI in mobility in the near term are driver assistance technologies and reusable software infrastructure. In turn, this enhanced software, when paired with increasingly capable high-powered compute, is the key enabler for higher levels of autonomy.

Future car: From software-defined to AI-defined

The auto industry is moving beyond software-defined vehicles toward AI-defined platforms, where *intelligence* rather than code shapes functionality and differentiation. Why? It enables continuously intelligent, data-driven automotive platforms. This adaptability is becoming core to the driving experience because it can be changed and upgraded over time – and not limited to when the car is manufactured. But the trickle-down effect is real: this shift is rapidly impacting vehicle manufacturers, suppliers and technology providers through hardware and software needs required to enable this shift.

The implication? Vehicles are becoming part of a broader physical AI ecosystem linking production, energy and operations into a continuous automated loop.

Start with software, end in autonomy

Autos offer something no other industry can match at scale: huge production volumes, strict safety requirements and complex operating environments. All of this makes cars the ideal place to mature and industrialize embodied intelligence long before full autonomy becomes mainstream.

Vehicle intelligence is no longer just an "autonomous driving feature" – it's evolving into a core layer of *physical AI infrastructure* built directly into the vehicle. This intelligence now includes engineering tools, safety-critical operating systems and in-vehicle AI software – not just driving policy. Once developed, these capabilities can be reused across different models, brands and even entirely different types of machines. This shift mirrors the broader physical AI landscape, where companies are racing to create intelligence that can be certified, scaled and transferred across domains.

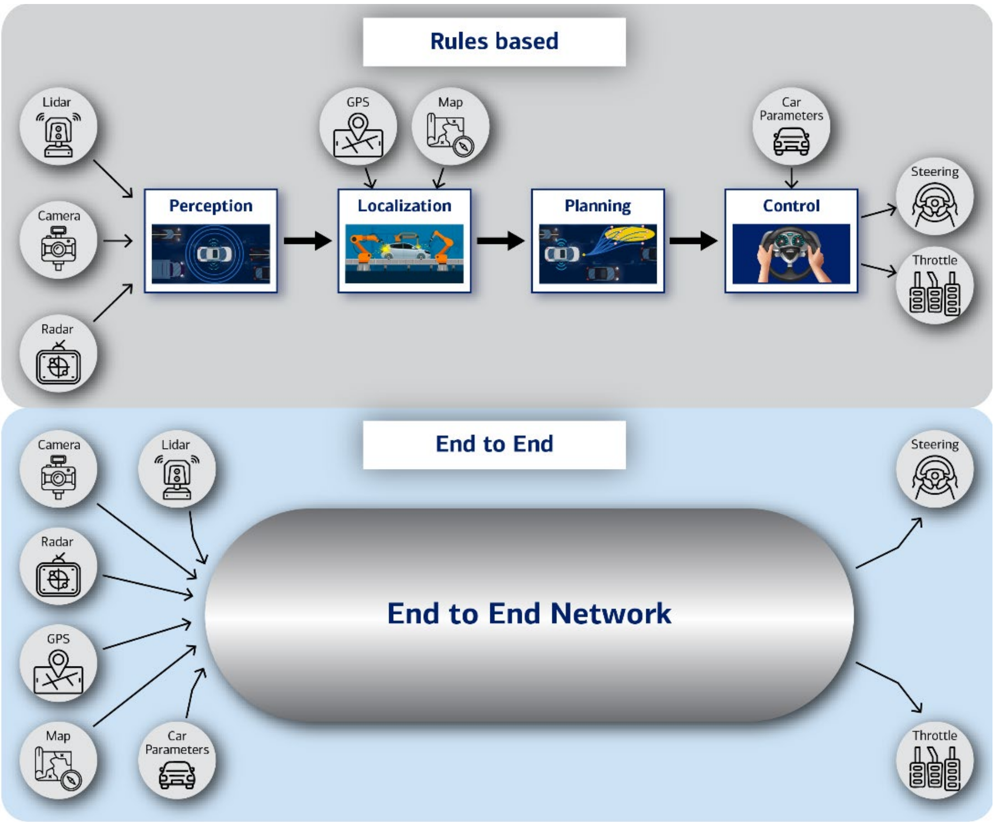
But physical AI isn't just software magic – it's constrained by real-world engineering and hardware realities. Early choices in sensors, compute and wiring set the ceiling for future autonomy and define how much intelligence you can run on the hardware the vehicle ships with. Most cars can add new features over time, but they can't be retrofitted for higher autonomy levels if they lack sensors, compute or fail-safes.

Unlocking a shared platform

As mentioned in [Physical AI, part 1: The basics](), end-to-end AI could enable a shared intelligence platform across both robots and AVs, since the same approach can train systems to manipulate objects or navigate roads (Exhibit 1). This unification promises faster rollout, lower costs and more scalable deployment than traditional modular methods, according to BofA Global Research. However, it also introduces trade-offs – requiring far more data and compute and creating tougher safety-validation challenges because errors are harder to trace – prompting companies to add extra models or software layers to compensate.

Exhibit 1: AV deployments could accelerate if the tech generalizes to new markets with embodied intelligence

Infographic depicting the shift from rules-based to end-to-end AI single-neural network



Source: BofA Global Research

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New revenue, new markets

Software-defined vehicles unlock new revenue streams: ongoing feature upgrades, services and fleet-level optimizations delivered over the air. A central compute "brain" orchestrates these capabilities. And the benefits don't stop with passenger cars. According to Applied Intuition, roughly 90% of what's needed for safety-critical physical AI applies across trucks, industrial machines, mining equipment and defense systems. Autos effectively *fund* the ecosystem for physical AI everywhere else.

Why do we need autonomous vehicles?

Beyond the economic rationale, road safety and a range of demographic challenges underpin the need for AVs:

- **Safety**: Around 94% of road accidents stem from human error, making safety a structural limit of manual driving.[1]
- **Structural labor shortage**: A global driver shortage is emerging as workforces age and fewer people enter the profession. The global shortfall of roughly four million truck drivers is expected to double by 2028. And in Europe, for instance, the average truck driver is 47 and the average bus driver is 50 – both older than the average working age of 43.
- **Regulatory constraints**: Hours-of-service rules cap driving time, reducing effective capacity even where drivers are available. The European Union (EU) and United Kingdom (UK) limit total driving at nine hours a day with the possibility to extend to 10 hours a day twice a week, with a maximum of 90 hours permitted over two consecutive weeks. Regulators in

[1] Occupational Road Safety Alliance. (n.d.). *Reducing Driver Error Accidents*.

Japan capped working hours and overtime in 2024 equivalent to ~6% of driver hours, exacerbating the already limited driver supply.

- **Rising transport costs**: Higher wages, fuel, tolls, insurance and compliance costs are pushing freight costs structurally higher.

- **Supply and demand imbalance**: Freight demand continues to grow (notably e-commerce), while capacity is constrained by labor and regulation.

- **Low asset utilization**: Roughly 20-30% of long-haul miles are run empty, highlighting inefficiencies AV fleets can address.

- **System resilience**: AVs reduce sole dependence on human availability, improving reliability amid labor volatility and disruptions.

Rise of the robotaxis

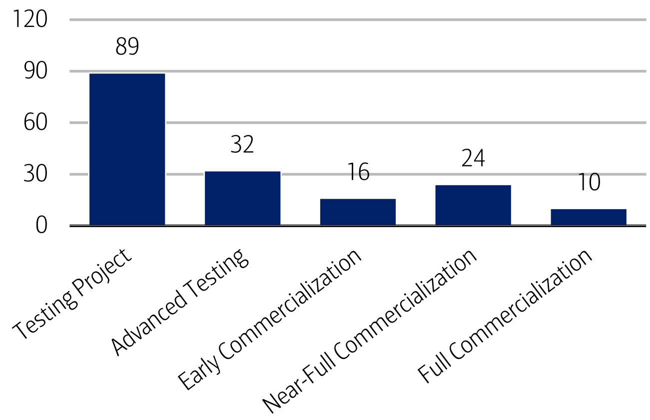
Robotaxi commercialization is quickly becoming a reality, with more than 170 active robotaxi deployments globally (Exhibit 2) – 40% in US and 24% in China (Exhibit 3). There are now 10 cities with fully commercial operations, including Atlanta, Austin, Los Angeles, Miami, Phoenix and San Francisco, in the US. These operations are defined as operating on public roads, accepting passengers, charging a fee, fully driverless without a safety driver and able to operate all day in any weather. Many more will likely follow, with 24 nearing commercialization (meeting four of these five criteria), and 16 at an early commercial stage (three of the criteria).

However, scaling is a challenge. Vehicles with sufficient sensors and capabilities remain expensive. Regulation is fragmented. Edge cases – the very rare and hard-to-predict road incidents – remain an engineering challenge to solve for and develop a sufficient safety case. But with the recent breakthroughs in AI and compute, the AV industry has a better opportunity to break through these bottlenecks and accelerate the rollout, evident from the first commercial operations beginning to expand.

But why now? Various hardware costs have fallen by more than 50% compared to previous AV models. And removing the driver could halve the cost per mile, unlocking more addressable markets. For more on how robotaxis work and the underlying technology stack, refer to The road ahead: The future of autonomous vehicles.

Exhibit 2: There are 171 active robotaxi deployments, 10 of which are fully commercial, with 40 more in early or near-full commercialization

Number of active robotaxi deployments and stage of commercialization



Source: BloombergNEF as of March 4, 2026, BofA Global Research

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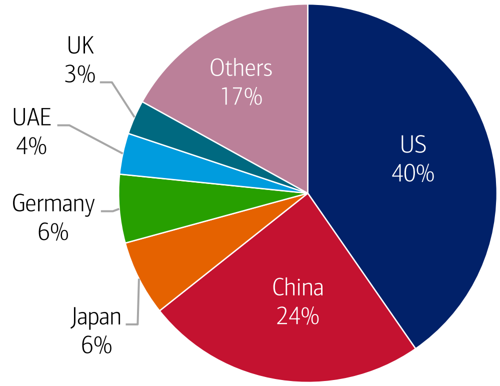
Exhibit 3: The US has the most active robotaxi and shuttle deployments to date (69, or 40% of the total)

Robotaxi and shuttle deployments globally (%)



Source: BloombergNEF as of March 4, 2026, BofA Global Research

Note: UAE = United Arab Emirates, UK = United Kingdom

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Ride-hail platforms racing to announce AV partnerships

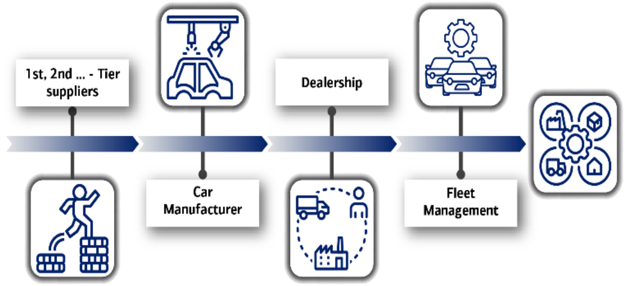
Global ride-hail platform leaders are increasingly partnering with, or investing in, AV developers to secure future access to driverless fleets, rather than build the technology themselves. Strategically, per BofA Global Research, AVs offer ride-hail operators greater control over service reliability and pricing, reduced dependency on human drivers and a long-term path to margin expansion, while allowing AV companies to scale faster by tapping into established global demand networks.

AV tech could blur the lines between private and shared vehicles

Multiple autonomous operating models are emerging in parallel – from privately owned AVs to shared robo-shuttles and ride-hailing robotaxis – and their boundaries may eventually blur. Today's mobility ecosystem is anchored in a traditional auto value chain centered on manufacturing, selling and maintaining vehicles (Exhibit 4). As AVs become commercially viable and lower the cost per mile, they could catalyze a shift away from personal car ownership toward on-demand mobility services. In this model, privately owned AVs might serve personal travel while also joining shared fleets when idle, and new tech-enabled platforms would make it easy for customers to choose the right mode for each trip. As a result, fleet management and multimodal aggregation could become increasingly central and lucrative within the future mobility value chain (Exhibit 5).

Exhibit 4: The current auto value chain is centered on manufacturing, selling and maintaining vehicles

Infographic illustrating current auto value chain



Source: BofA Global Research

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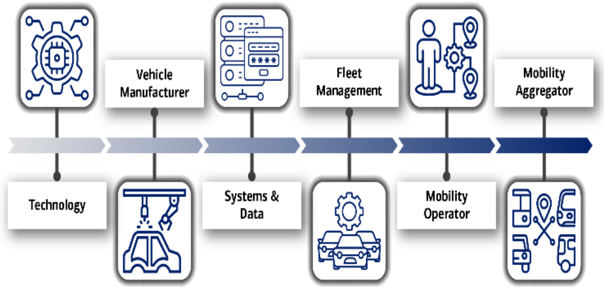
Exhibit 5: The commercial readiness of AVs could revolutionize the value chain, with fleet management and aggregation of other transportation services playing a more lucrative role

Infographic illustrating the potential value chain for (auto) mobility



Source: BofA Global Research

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Expanding AV business models, a $740 billion addressable market

As AV technology scales, developers are increasingly pursuing hybrid business models, combining fleet ownership, software licensing and partnerships rather than a single go-to-market approach. Reflecting this flexibility, AV software and systems could represent a $740 billion addressable market across consumer and commercial vehicles, according to Wayve.

Keep on (autonomously) trucking

Autonomous freight networks (AFNs) aim to disrupt the $4 trillion freight trucking market, creating a structure that can save significant operational costs by removing drivers from the cab, with enhanced fuel efficiency and routing compared to traditional truck operating models. The core technology is similar to that being added to passenger cars – a combination of sensors and software – albeit with different design, integration and operational design domain (ODD). For example, most AV car projects thus far focus on urban driving, whereas AV truck projects focus on highways and longer distances. This overlap is accelerating the emergence of physical AI, where core perception, planning and control infrastructure can be shared across autonomous platforms while being tuned for specific use cases.

AVs going commercial in trucking, delivery and logistics

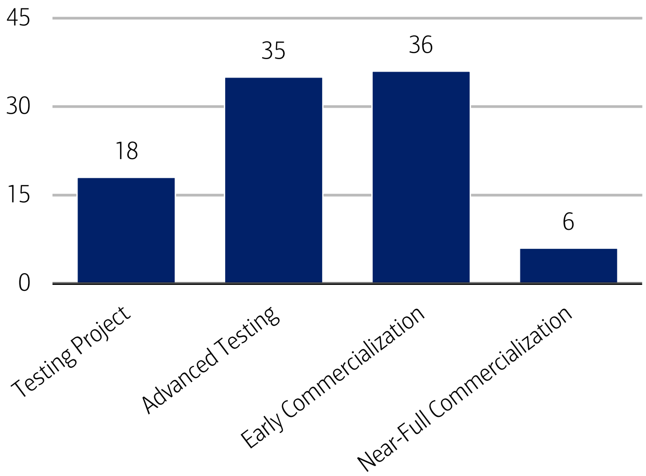
As seen in Exhibit 6, there are currently 95 operational commercial vehicle AV projects, of which, 42 are at the early or near-full commercialization stages, measured across five metrics:

1. **Operating on complex routes**: Routes either cross state/country boundaries or require both highway and street driving, validating the ability to handle diverse driving scenarios.
2. **Manufacturer contracts**: Secured contracts with vehicle manufacturer(s), a pathway towards economies of scale and cost efficiency.
3. **Freight provider contracts**: With freight brokers, shippers or carriers, signaling AV developer ability to launch delivery or freight services and generate revenues.
4. **Fully driverless**: Operate without safety drivers, indicating AV tech is ready for deployment without human intervention.
5. **All-time/weather**: Able to operate at all times of the day and in all weather conditions.

No provider meets all criteria yet, but six of the projects meet four, meaning they're at "near-full commercialization." Overall, most deployments are in the long-haul trucking space, likely owing to the easier testing opportunity on highway routes and longer distances (Exhibit 7). According to McKinsey, commercial launches are expected to start in 2027.

Exhibit 6: Of the 95 operational commercial AV projects, 42 are at an early or near-full commercialization stage

Trucking and logistics AV deployments



Source: BloombergNEF as of March 4, 2026, BofA Global Research

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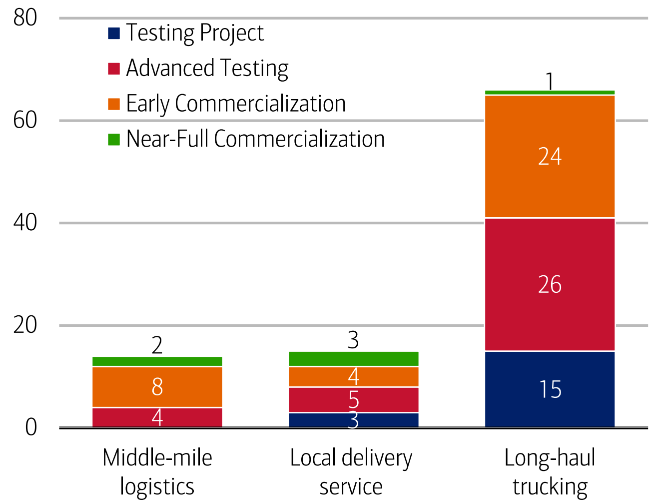
Exhibit 7: There are 66 ongoing long-haul AV truck projects, 25 of which are at some commercial stage

Commercial AV categories with stage of commercialization



Source: BloombergNEF as of March 4, 2026, BofA Global Research

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$4 trillion road freight total addressable market (TAM)

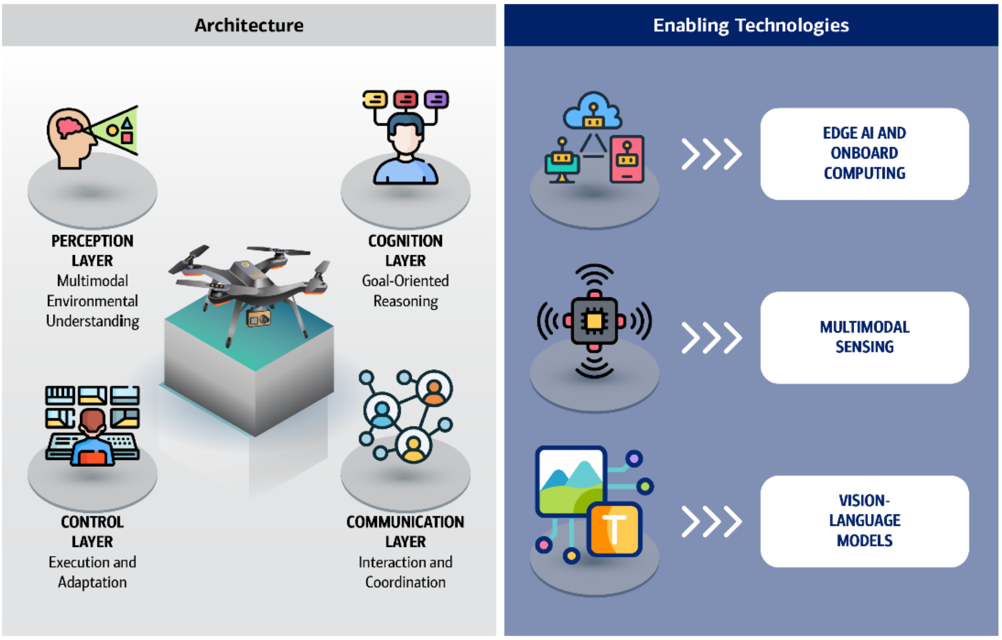
Global freight trucking is a $4 trillion annual revenue industry. Asia is the largest market ($1.7 trillion), followed by the US ($0.8 trillion). AV trucks could create over $40 billion in value by 2035 in the US alone, per BofA Global Research – assuming a 10% share of freight miles using AV trucks – a benefit shared between freight customers, freight companies, autonomous software/system developers and truck original equipment manufacturers (OEMs).

Drones: Physical AI is about to take off

Unlike traditional automation or cloud-based AI, drones require continuous, real-time integration of perception, decision-making and control on constrained onboard hardware (Exhibit 8). Latency, power and safety limits force intelligence to operate fully at the edge, with no reliance on persistent connectivity or human intervention. These constraints mirror those faced by AVs, robots and industrial machines, positioning drones as a leading testbed for embodied autonomy rather than a niche application.

Exhibit 8: A combination of advanced AI models, sensing and compute enable drones to operate autonomously

An infographic illustrating autonomous drone architecture and enabling technologies



Source: BofA Global Research

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Highways in the sky enabled by AI and hardware

In order to fly, drones combine robotics and aeronautics technologies, using a mix of propellers, sensors and flight controllers to maintain stability and control. Improvements in onboard compute, perception and navigation systems increasingly enable drones to perform entire missions autonomously, using a combination of GPS (global positioning system), cameras and sensing technologies such as radar and LiDAR (light detection and ranging) – paralleling the technology stack used in AVs on the ground.

Amenable regulation: Beyond visual line of sight (BVLOS)

Despite advances in autonomy, drone delivery was historically constrained more by regulation than by technology. In most markets, commercial drones were required to remain within the visual line of sight of a human operator (BVLOS restrictions), effectively limiting range, fleet size and economic viability. This requirement tied each flight to active supervision, adding labor costs and preventing scalable delivery networks.

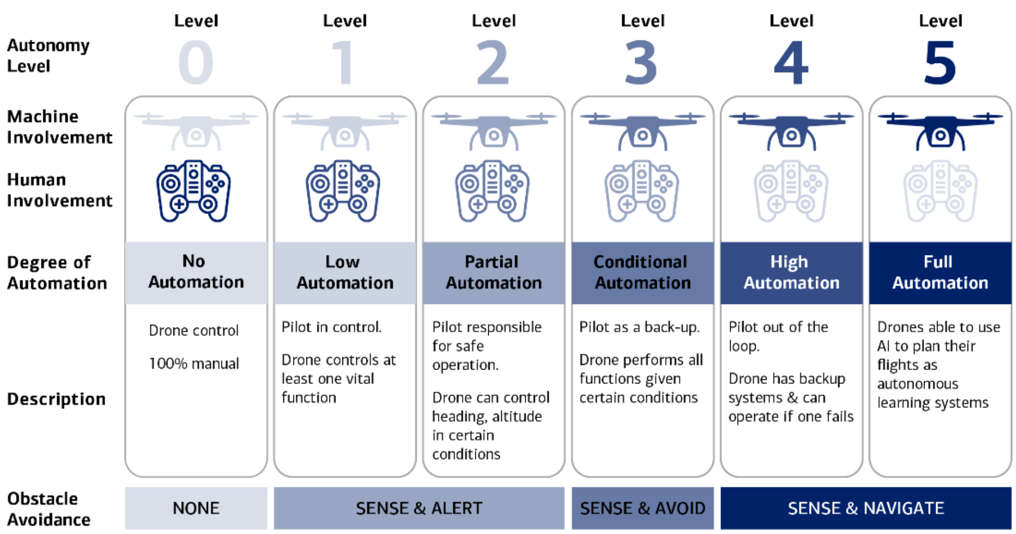
Gradual regulatory change has begun to relax these constraints. Authorities are increasingly permitting drones to operate outside an operator's direct field of view, provided they can safely plan routes, detect obstacles and remain within predefined operating areas. This shift enables end-to-end autonomous routes, hub-and-spoke delivery models and fleet-level supervision, forming the regulatory foundation for early commercial drone networks.

Just like AVs: Increasing levels of drone autonomy

As with on-road AVs, drones are increasingly classified by levels of autonomy, ranging from limited automation to highly autonomous operation under defined conditions (Exhibit 9). Most commercial deployments have progressed to Level 3 (conditional automation), where drones navigate and avoid obstacles independently, with remote pilots available for intervention. Meanwhile, early Level 4 deployments are emerging in geofenced and tightly constrained environments, enabling largely end-to-end autonomous flight. These systems are being commercialized across mapping, inspection and early delivery use cases, while full Level 5 autonomy remains out of reach due to ongoing reliance on structured airspace and human supervisory oversight.

Exhibit 9: Increasing levels of drone autonomy are driven by tech and regulation

Infographic depicting levels of drone automation



Source: Drone Industry Insights, BofA Global Research

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Drones hit the sky: Delivery, defense and public safety

Drones are now crossing into commercial deployment across several use cases, particularly in the end markets of delivery, defense and public safety. While each market is driven by different demand dynamics, all rely on the same core physical AI capabilities – edge autonomy, real-time planning and control, fleet coordination and safety-critical validation – reinforcing drones' role as both a proving ground and an operational entry point for embodied AI.

Taken together, these developments point to a structural shift. Drones are forcing governments to rethink how they control airspace, protect infrastructure and manage risk – both in conflict and at home. The result is not just a new category of vehicles, but a new layer of digital and physical infrastructure governing anything that moves through the air.

Commercial drone market: $41 billion in 2025, projected $57 billion in 2030

The commercial drone market (excluding military applications) has already reached meaningful scale, with revenues growing from around $34 billion in 2023 to approximately $41 billion in 2025 and projected to reach $57 billion by 2030.[2] The largest use case thus far is mapping and surveying – particularly in energy, construction and infrastructure – followed by inspection and filming.

Advances in AI-enabled perception, navigation and planning are driving deployment across multiple sectors with commercial applications expanding into agriculture, warehouse logistics and last-mile delivery as tech advances and regulation matures. In agriculture, drones enable precision imaging and automated spraying. In construction and infrastructure, they support high-resolution mapping, digital twin generation and automated aerial surveys. In public safety, drones provide rapid situational awareness and remote assessment, increasingly acting as first responders in time-critical incidents.

When military applications are included, the market is substantially larger. IDTechEx estimates the total global drone market will reach ~$69 billion in 2026, with defense as the single largest revenue segment, reflecting the higher unit costs and system complexity of military platforms. Looking ahead, they project the total drone market could approach ~$150 billion by 2036.

Defense tech: Autonomy reshaping modern warfare

Drones are reshaping modern warfare by delivering outsized impact at low cost. Recent conflicts have demonstrated how commercially derived drones can be rapidly deployed for surveillance, targeting and disruption, often at a fraction of the cost required to defend against them. This asymmetry is shifting military focus away from more sophisticated drones toward detection, tracking and low-cost neutralization.

What are unmanned aircraft systems?

Unmanned aircraft systems (UAS) are now widely integrated into modern airspace, particularly in military operations where they provide intelligence, surveillance, reconnaissance and strike capabilities without putting human crews at risk. In everyday usage, the term "drone" is often used loosely, but it obscures an important distinction: a UAV (unmanned aerial vehicle) refers only to the unmanned aircraft itself, while a UAS includes the full operating system, such as ground control stations, communications links, sensors, software and human operators. Most systems today remain remotely piloted, though advances in autonomy are increasingly enabling drones to operate with reduced human involvement.

From airspace control to public safety

As drones proliferate, low-altitude airspace is becoming a contested domain around airports, stadiums, borders and critical infrastructure. Governments are responding by treating airspace as actively managed infrastructure, supported by digital identification, monitoring and enforcement systems. Regulatory frameworks enabling routine BVLOS operations are also laying the groundwork for both commercial scale and security oversight.

Drone security is also converging with policing and emergency response. Drones are increasingly used by police and fire departments as "first responders" to assess incidents before personnel arrive. At the same time, those same agencies must be able to prevent unauthorized drones from interfering with emergencies, prisons or public gatherings. This is driving demand for integrated platforms that combine airspace monitoring, drone operations, evidence capture and incident management into a single system – blurring the traditional boundary between defense technology and public safety software.

[2] Drone Industry Insights. (2025, March). *Drone Market Report 2025-2030*.

Contributors

Vanessa Cook

Content Strategist, Bank of America Institute

Lynelle Huskey

Analyst, Bank of America Institute

Sources

Martyn Briggs

Equity Strategist, BofA Global Research

Haim Israel

Equity Strategist, BofA Global Research

Felix Tran

Equity Strategist, BofA Global Research

Lauren-Nicole Kung

Equity Strategist, BofA Global Research

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Transformation

Physical AI, part 2: Humanoid robots

12 March 2026

Key takeaways

- AI is redefining robotics, enabling a new generation of humanoid robots that can perceive, learn and act autonomously in the physical world. After decades of incremental progress, the field has reached a turning point, with breakthroughs in generative AI and large language models accelerating the rise of "embodied intelligence" and reshaping expectations for what service robots can accomplish.
- The humanoid robot population is poised for take-off. BofA Global Research projects that the number of humanoid units in operation could reach three billion by 2060, surpassing cars on a per-capita basis. While industrial and service applications are expected to lead adoption in the near term, household humanoids are projected to accelerate quickly and ultimately account for the largest share - 62% - by 2060.
- In part 2 of our physical AI series, we outline what you need to know about humanoid robots - including how they work, how they're being used today and key challenges the industry faces as it moves toward large-scale deployment.

Humanoid robots 101

Humanoid robots are a type of service robot designed to move, perceive and interact in ways that resemble human beings. As a type of physical AI, they embody artificial intelligence in a dynamic, mechanical form capable of acting on the world (for readers new to this topic, Physical AI, part 1: The basics provides a helpful foundation). This enables them to operate in complex, unstructured environments, rather than controlled settings. Unlike traditional service robots, which are built for narrow, repetitive tasks, humanoids are required to navigate real-world spaces, interpret human behavior and respond through physical action – placing far greater demands on AI for sensing, motion control and natural-language interaction.

After decades of incremental progress, the field reached a turning point in 2023, as breakthroughs in generative AI and large language models (LLMs) accelerated the emergence of “embodied intelligence.” By giving AI a physical form, humanoid robots can learn directly from the world, translate language into action and adapt to people in real time – reshaping expectations of what robots can do.

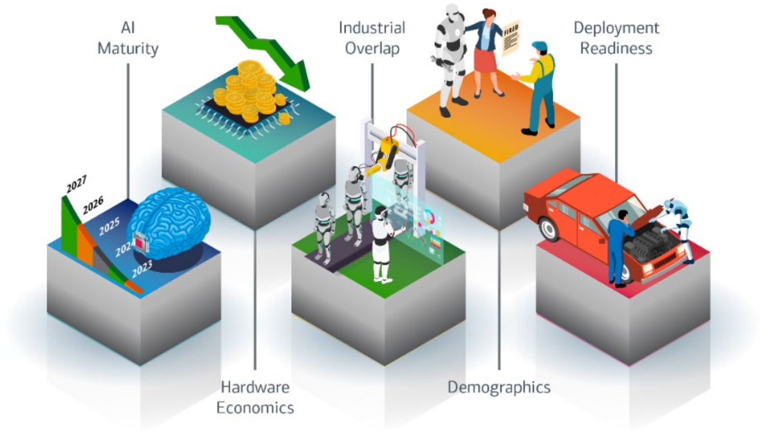
Humanoid adoption is accelerating – but why?

According to BofA Global Research, humanoid robot adoption is speeding up (Exhibit 1), mainly because of:

1. **AI maturity**: Model breakthroughs are now enabling more general-purpose learning, better long-term reasoning and more reliable AI operation in unstructured, real-world human environments.
2. **Hardware economics**: Rapid cost declines and improvements across key humanoid components (e.g., batteries, motors, sensors, actuators and onboard AI compute) have lowered the required bill of materials (hardware costs) and boosted overall feasibility of humanoids.
3. **Industrial overlap**: Humanoid robots can tap into existing supply chains, manufacturing methods and capital investment from electric vehicles (EVs), autonomous tech and AI hardware sectors, helping them scale faster and reduce execution risk.
4. **Demographics**: Ongoing labor shortages, aging workforces, wage inflation and high turnover make flexible robotic labor economically appealing across industries, such as manufacturing, logistics and services, even before humanoids fully match human abilities.
5. **Deployment readiness**: Humanoid form factors fit with existing tools, buildings, and workflows, which cuts down on deployment barriers. Together, improved safety, autonomy and self-learning reduce integration, regulatory and operating friction.

Exhibit 1: Tech breakthroughs, economics and demographics are key drivers behind humanoid adoption

Infographic illustrating factors accelerating humanoid robot adoption



Source: BofA Global Research

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Scaling-up from pilot to production

The robot rush: Increased investment is happening now

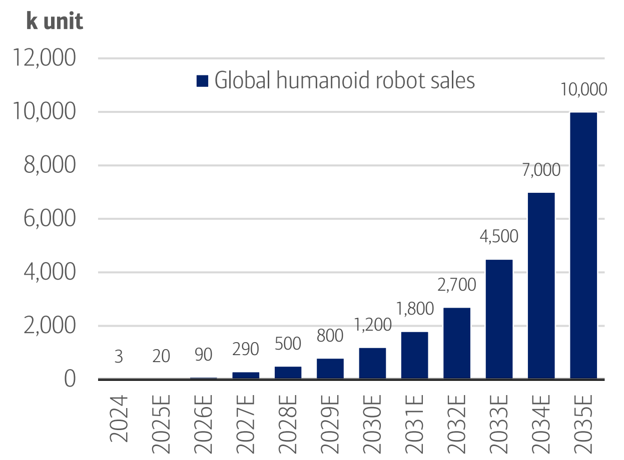
The commercial race to develop humanoids is heating up as the tech matures and prices decline. Investments have risen sharply – from $0.7 billion in 2018 to $4.3 billion in 2025.[1] And as of January 2026, over 50 companies were in the process of building their own humanoids,[2] with 150 commercial launches to date.[3] So far, the development landscape has bifurcated between companies vertically integrating robots in their own factories and those selling or renting humanoids to their customers.

A robot population boom is on the horizon

BofA Global Research forecasts annual humanoid robot shipments will surge from 20,000 units in 2025 to 10 million by 2035, implying an 86% compound annual growth rate (CAGR) (Exhibit 2). In 2026 alone, sales are expected to reach 90,000, as leading companies ramp up production capacity and scale deployments.

Exhibit 2: Annual humanoid robot shipments are projected to reach 1.2 million in 2030 and 10 million by 2035

Annual global humanoid robot shipments (thousands (k))



Source: BofA Global Research

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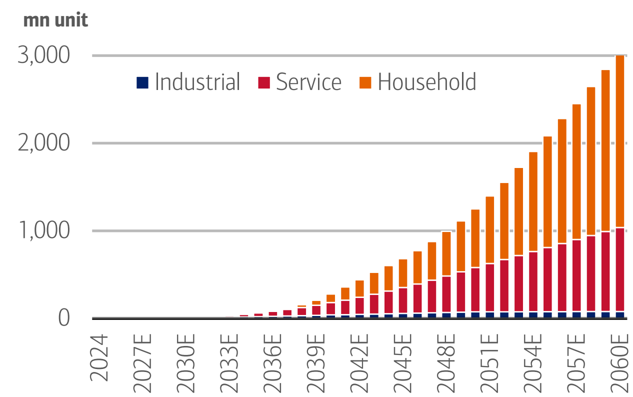
Exhibit 3: The humanoid robot "population" could reach three billion by 2060E

Long-term forecast of humanoid robot units in operation (UIO) by application (millions (mn))



Source: BofA Global Research

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[1] Humanoid Guide. (2026). *Humanoid robot guide*.
[2] Botfino. (n.d.). *Humanoid Robotics Company Intelligence*.
[3] Humanoid Guide. (2026). *Humanoid robot guide*.

As seen in Exhibit 3, the global "robot population" could reach 300 million by 2040 and even three billion units by 2060 – outnumbering cars per capita. While analysts believe industrial and service applications are likely to come first, by 2060 they project household humanoids will make up the largest share (62%, or two billion units).

Economies of scale and tech breakthroughs will significantly reduce costs

Despite recent improvements, the total bill of materials (BOM) or hardware cost of humanoid robots remains relatively high, with pilot-stage development ranging from $90,000 to 100,000 per unit. However, standardizing product specifications has already brought costs down considerably, with further reductions possible with economies of scale. For example, BofA Global Research estimates a built-in China humanoid robot's BOM cost was $35,000 in 2025 but expects it to be roughly cut in half to below $17,000 by 2030, driven by scale effects and continued improvements in component design.

How do humanoid robots work?

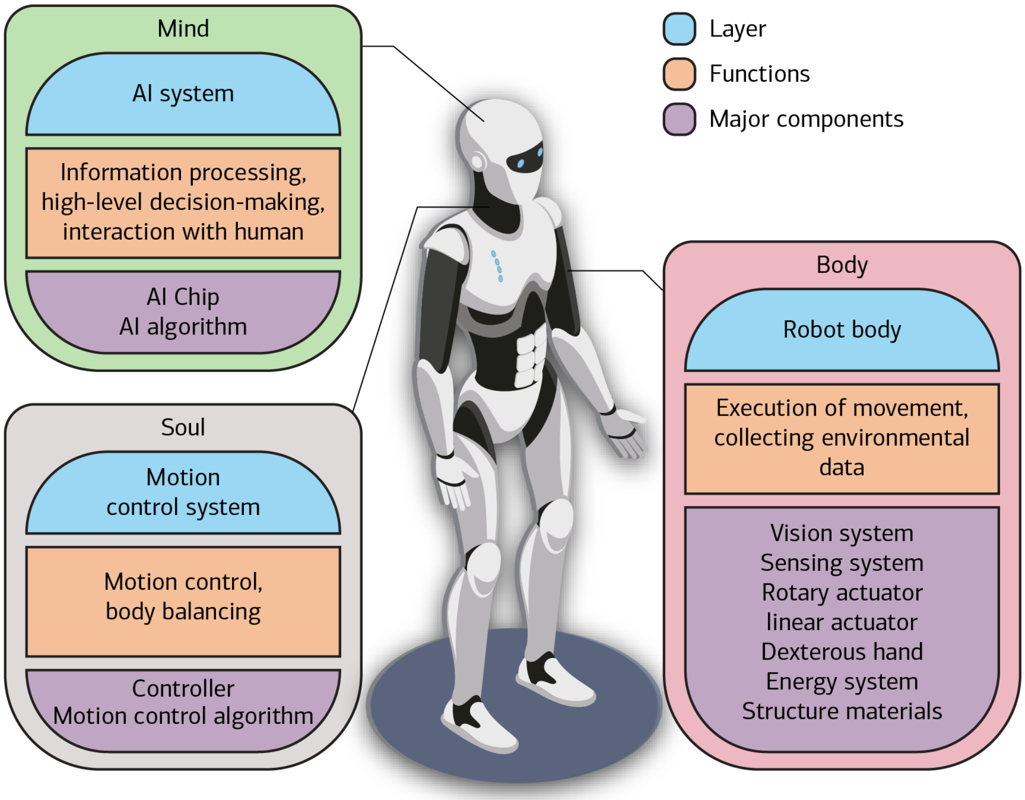
Humanoid robots operate by combining central AI control and compute with mechanical components – actuators, motors, reducers, hands and sensors – to turn intelligence into physical action. While AI systems perceive and decide, high-performance compute processes information in real time, and tightly integrated actuators and sensors deliver balance, dexterity, strength and safe interaction. Together, these elements form the foundation of capable, scalable humanoid autonomy.

Integrating the "mind, body and soul"

The structure of a humanoid robot can be framed as the hardware and software that collectively create the "mind, body and soul" (Exhibit 4). The mind refers to the AI system, which handles high-level cognition, such as environmental understanding, task planning, model inference and interaction with humans. The body of the humanoid comprises the physical components, including vision and sensory systems, actuators, dexterous hands, energy systems and structural materials. And last, the soul represents the motion control system – the robot's internal coordinator – responsible for balance, movement planning and motion execution.

Exhibit 4: A combination of AI chips and algorithms make up the humanoid's mind and soul, with multiple components required for the body – such as actuators, motors, batteries and sensors for vision

Infographic illustrating the mind, body and soul of humanoid robots



Source: BofA Global Research

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Bot mechanics

From components to capability

Humanoid robots are built from a tightly integrated combination of computing, actuation, sensing and mechanical components that together enable perception, movement and interaction in real-world environments (Exhibit 5). Rather than operating independently, these elements must be synchronized in real-time, with performance evaluated by how effectively intelligence, motion and hardware constraints are integrated. For a more in-depth discussion of humanoid components, check out our prior publication: Humanoid robots 101.

Exhibit 5: Advances in AI and motion control, combined with precision hardware, such as actuators and motors, are the core tech embedded in humanoid robots

Infographic illustrating key components in a humanoid robot

Component		What it does	Why it matters
Control System (AI + Motion Control)		Interprets the world, decides what to do, and coordinates how the robot moves.	Determines how intelligent, safe, and adaptable the robot is in real-world, human environments.
AI Computing & Chips		Provides the processing power needed to run perception, reasoning, and control in real time.	Sets the ceiling on capability, responsiveness, and autonomy; critical for moving from demos to deployment.
Rotary Joint Actuators		Drive rotational movement in joints like shoulders, elbows, hips, and wrists.	Core to mobility and manipulation; quality of integration determines smoothness, precision, and noise.
Linear Joint Actuators		Enable extension and compression in joints such as knees, ankles, and arms.	Essential for strength, balance, and lifting; heavily influences stability and payload capacity.
Dexterous Hands		Allow the robot to grasp, hold, and manipulate objects, including delicate items.	Unlocks household, service, and fine assembly tasks; one of the hardest and most valuable capabilities to replicate.
Reducers (Mechanical Gearing)		Convert motor speed into usable torque for precise joint movement.	Critical for accuracy, durability, and safety; a key differentiator between industrial-grade and prototype robots.
Torque Motors		Deliver compact, high-power movement directly at the joints.	Enable lighter, faster, and more responsive robots with fewer mechanical compromises.
Sensory System (Touch, Force, Balance)		Helps the robot feel contact, understand forces, and maintain balance.	Enables safe interaction with people

Source: BofA Global Research

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Achieving higher degrees of freedom

Advances in dexterous robotic hands are expanding what humanoids can manipulate in real-world settings. A key driver is higher degrees of freedom (DoF) – the number of independent axes of movement across a robot's joints – which directly determines how precisely and flexibly it can position its limbs, hands and fingers. These gains are amplified by fully rotational joints that can turn continuously through 360°, rather than stopping at human-like anatomical limits.

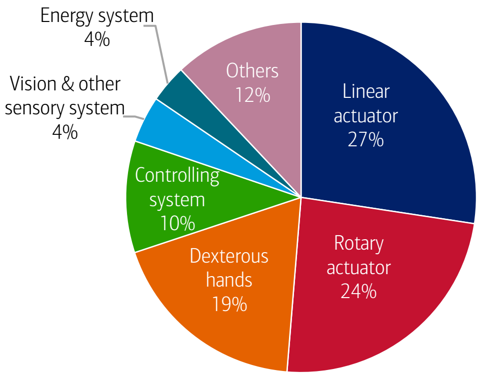
Alongside this, compliant actuators – motors designed to yield under force rather than remain rigid – allow robots to absorb impacts, regulate force and adapt safely when grasping or interacting with uncertain objects. Combined with tactile-sensing and learning-based control, higher DoF and compliance materially increase *usable* dexterity. However, according to BofA Global Research, progress remains constrained by limited high-quality manipulation data and the difficulty of replicating human-level dexterity and improvisation.

Bill-of-material economics: Actuators are actually expensive

The actuation system is the muscle of a humanoid robot, controlling movement, balance and force. The system integrates motors, drives, transmissions and sensors into compact modules that are deployed across both rotary and linear joints. A humanoid typically uses dozens of actuators, underscoring their central role in performance and cost.

By 2030, BofA Global Research projects that linear actuators – which enable extension and compression in humanoid joints – will account for the largest BOM share (27%), followed by rotary actuators (24%) and dexterous hands (19%) (Exhibit 6).

Exhibit 6: Actuators account for more than 50% of the total BOM cost
BOM cost breakdown (by module/system, 2030)



Source: BofA Global Research

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Humanoid use cases

In an aging, labor-constrained global economy, BofA Global Research believes humanoid robots could offer an increasingly cost-competitive form of physical labor, underpinned by improving AI autonomy and manufacturing scale.

From the factory floor to in-home assistants

Humanoid robots are being deployed in a growing range of applications (Exhibit 7). In manufacturing, automotive and warehouse environments, they handle tasks related to moving materials, inspection and assembly. In retail, hospitality and live events, they support customer interactions, restocking and front-of-house services. Emerging applications also include healthcare, homes and education, where humanoids are positioned for assistance, monitoring, training and learning support. Together, these developments reflect a gradual shift from structured industrial settings towards more human-centered environments.

Exhibit 7: From cars to caring – humanoids have a wide variety of use cases
Infographic illustrating humanoid robot use cases

Humanoid Robot Use Cases

Use Case		Description
Manufacturing		Performing industrial assembly, machine tending, and materials-handling tasks.
Warehousing & Logistics		Handling picking, packing, palletizing, and goods movement in distribution settings.
Automotive		Supporting assembly lines, component handling, and factory-floor operations in automotive production.
Retail & Hospitality		Delivering customer greeting, wayfinding, and front-of-house assistance.
Special-Purpose/ Events		Executing demonstrations, exhibitions, performances, and specialised showcase tasks.
Healthcare & Support		Carrying out non-clinical assistance, supply movement, and routine operational tasks in care environments
Home Assistance		Assisting with household chores, tidying, and simple object-handling tasks.
Education & Research		Supporting simulation, training, experimentation, and human-robot interaction studies.

Source: BofA Global Research

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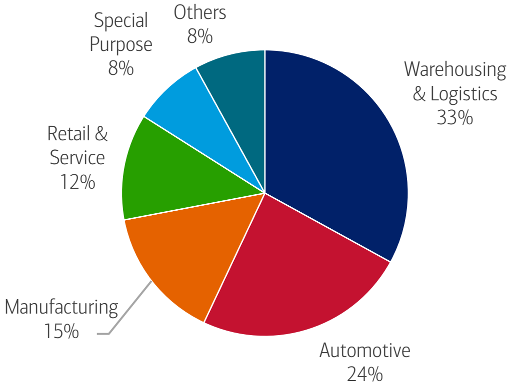
So far, initial humanoid use cases have been mostly industrial and are expected to remain that way in the near term. By 2027, Counterpoint Research estimates that 72% of installations will be in logistics, manufacturing and automotive (Exhibit 8).

Exhibit 8: Initial humanoid use cases are industrial
Humanoid use cases in 2027, by industry (%)



Source: Counterpoint Research, BofA Global Research

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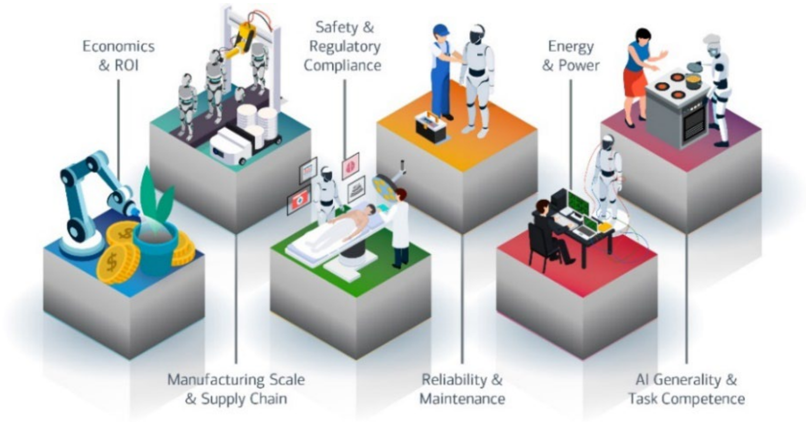
It's tough being a humanoid

As manufacturing volumes of humanoids increase, the challenges facing companies building and deploying them are shifting from production to techno-economic – achieving the levels of performance, reliability and cost that make sense commercially. The challenges to scaling humanoids, according to BofA Global Research, are (Exhibit 9):

- **Economics and return on investment (ROI):** High unit costs, uncertain labor substitution benefits and expensive repairs make it difficult to achieve compelling payback periods.
- **Manufacturing scale and supply chain maturity:** Some humanoid parts, like actuators, sensors, reducers and batteries, lack the volume, reliability and cost structure needed for large-scale humanoid production.
- **Safety, certification and regulatory compliance:** Standards for safe human-robot co-working are still emerging, slowing approvals for real industrial deployment.
- **Reliability, uptime and maintainability:** Current humanoids remain too fragile and maintenance-intensive for 24/7 industrial environments that demand high uptime and durability.
- **Energy and power constraints:** Limited battery life and immature charging or hot-swap ecosystems restrict continuous operation and reduce productivity.
- **AI generality and task competence:** Autonomous systems still struggle with unstructured, variable, real-world tasks, limiting the economic usefulness of humanoids today.

Exhibit 9: Challenges remain ahead of commercial operation of humanoid robots at scale
Infographic illustrating key challenges behind scaling humanoid robots



Source: BofA Global Research

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Contributors

Lynelle Huskey

Analyst, Bank of America Institute

Vanessa Cook

Content Strategist, Bank of America Institute

Sources

Martyn Briggs

Equity Strategist, BofA Global Research

Haim Israel

Equity Strategist, BofA Global Research

Felix Tran

Equity Strategist, BofA Global Research

Lauren-Nicole Kung

Equity Strategist, BofA Global Research

Disclosures